Thermodynamics and Statistical Mechanics

Lecture Notes

V. Balakrishnan

(transcribed from handwritten notes by Vishal Rao)

Contents

1 Lecture 20: Classical Statistical Mechanics **3**

2 Lecture 21 **9**

3 Lecture 22 **14**

4 Lecture 22 **17**

5 Lecture 23 **19**

6 Lecture 24 **24**

7 Lecture 25: Connection with Thermodynamics **30**

8 Lecture 25 (continued) **33**

9 Lecture 25 (continued): Criterion for Classical Statistics **36**

10 Lecture 26 **37**

11 Lecture 27: Phase Transitions **38**

12 Lecture 27 (continued) **40**

13 Lecture 27 (continued): Mechanical Example of Hysteresis **42**

14 Lecture 28 **42**

15 Lecture 28 (continued): Interactions and Ferromagnetism **45**

16 Magnetism and Phase Transitions **46**

16.1 Magnetization Calculation (General 3-D Case) 46

16.2 Langevin Function and Susceptibility 48
16.3 Two-Dimensional Case 49
16.4 Weiss Molecular Field Theory 50

17 Lecture 29: Ferroelectric Transitions and the Critical Region 51
17.1 Digression: Hysteresis and Domains 52
17.2 Family of Isotherms 52
17.3 Analogy with the Fluid Critical Point 53
17.4 Curie Temperature from the Mean-Field Equation 54
17.5 Susceptibility at $H = 0$ 54
17.6 Critical Exponent β 55
17.7 Critical Isotherm and Exponent δ 56
17.8 Susceptibility Exponent γ and Landau Theory 58

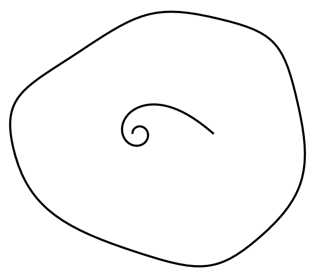
1 Lecture 20: Classical Statistical Mechanics

- Statistical mechanics helps in understanding the behaviour of systems which involve a large number of degrees of freedom (which are interacting non-linearly with one another).
- It is the science of averages. Also it is the study at the largest/longest length scales and time scales.
- Statistical mechanics is partitioned into two parts:
  - Equilibrium statistical mechanics.
  - Non-equilibrium statistical mechanics.
- Equilibrium statistical mechanics arose from the question that: are there any steady states (equilibrium states) which the system will finally settle down to? If there is, how does it behave, and so on?
- Thermodynamics is the study of systems at their longest length scales and time-scales.
  - So all fluctuations which happen at very small length scales and time scales are neglected, and we look at things at very long time scales and very long length scales.

Let us start with the Hamiltonian of a system. The system contains many particles and is isolated from the rest of the universe.

Isolated system

(exchange of matter & energy are prohibited).



$H(q,p)$

If the rules are time-translation invariant, then we know that the Hamiltonian is a constant of motion. $\left(\dfrac{dH}{dt}=0\right)$

$$H(q,p)=E \quad \text{(total energy of system).}$$

{with uncertainty δE}

Phase space becomes $6N$-dimensional for N particles. So we need to know $3N$ constants of motion which are in involution with each other. $\{C_i, C_j\}=0$ if $i \neq j$ [**?**]

Constants of motion will be:

1. H 1

2. total linear momentum (P) 3
3. Total angular momentum (L) 3
4. C.O.M position ($\vec{R}$) 3

Total 10 c.o.m [constants of motion].

To specify the state of the system we need a point in phase space $(q_1, q_2, \ldots, q_N,\, p_1, p_2, \ldots, p_N)$.

And that point will move on some $6N$-dimensional hypersurface.

- Due to lack of a sufficient number of constants of motion which are in involution with each other, we know that the differential equation of motion is not integrable, and the system of such kind is going to show chaotic behaviour with some positive Liapunov exponents. It is likely to wander around completely in a strange fashion, such that the system is ergodic on the energy surface.
- It does not have to be strictly ergodic; there can be a great deal of mixing (exponential separation of the volume element).

  Any way, exponential separation $\Rightarrow$ mixing $\Downarrow$ ergodicity (implies)

Ergodicity

- The volume element of phase space will come closer to every point of phase space in a given sufficient time. {definition of ergodicity}.
- The whole phase space is not accessible to the system because we specified its total energy.
- we become ignorant in knowing what happens to system at time $t \neq t_0$, and assume that every point/volume element of phase space is as equally likely as other points of phase space. We can term this principle as principle of maximum ignorance.

Fundamental Assumption/postulate of equilibrium statistical mechanics

$\rangle$ *"In a state of thermal eq^m, all accessible micro-states of the system are equally probable."*

"This single statement is enough to derive all equations of stat. mech, whereas thermal eq^m turns out to be a special case of it."

Thermal equilibrium

Two systems are said to be in thermal equilibrium if there is no heat flow between them when they are connected to each other.

A system is said to be in thermodynamic eq^m if it is in mechanical, physical and thermal equilibrium.

⟩ state of thermal eq^m also means that long time average of macroscopic variables reach a steady state and do not change with time. Such macroscopic quantities are Energy (internal), Total pressure, etc.

"All physical macroscopic quantities are time independent in a system of thermal eq^m state."

Note:-

⟩ Microscopic variables like velocity of a particle in certain d.o.f is not time independent, whereas the average velocity of all/each particles is time independent, which is zero, i.e. system is not moving anywhere.

⟩ The average values can only be independent of time if the probability distribution (over which the average is defined) is itself independent of time.

micro state:- Specifying the state of every constituent of the system. The set $(q_1 \cdots q_n,\, p_1 \cdots p_n)$ is the microstate of the system, or we can say that position in phase space is microstate, whereas the total momentum (avg) of all particles is macrostate.

By accessible microstate we restrict the velocity a particle can have, i.e. $H(q,p) = E_{\text{total}}$.

So a particle of such an isolated system can not have $K.E > E$. So some microstates are not accessible anymore.

"All those micro-states which are compatible with all restrictions applied on the system are termed as accessible micro-state."

"microstate tells what each and every particle is doing and macrostate is giving some gross/overall information."

If microstate energy is the only parameter of the microstate → (in quantum stat. mech.)

"If an isolated system has 10 J of total energy then we can have a single particle which has energy close to 10 J and rest of all are close to zero Energy. (microstate) This is equally probable as other states; followed from postulate of eq^m stat. mech."

In phase-space, probability density obeys

$$\frac{\partial f}{\partial t} = \{H, f\} \qquad \left(\begin{aligned} f &= \text{const. of motion} \\ \frac{df}{dt} &= \{f, H\} + \frac{\partial f}{\partial t} = 0 \\ \frac{\partial f_{eq}}{\partial t} &= \{H, f_{eq}\} \end{aligned} \right)$$

if probability distribution is independent of time

$$\frac{\partial f_{eq}}{\partial t} = 0 = \{H, f_{eq}\} \implies \text{implies } \underline{f_{eq} = f(H)}.$$

remember f_{eq} is not a physical observable.

$$f_{eq} = f_{eq}(H) = \delta(H(q,p) - E)$$

Since such an isolated system uses microstates, such an ensemble is termed as the microcanonical ensemble. A stat. ensemble with $f_{eq} = \delta(H(q,p) - E)$, which corresponds to an isolated system in thermal eq^m, is called the microcanonical ensemble (in phase space).

$\rangle$ If we assume we can not have a point in phase space but a volume shell of finite resolution, then only can we define the probability of occupying a certain state (volume element in phase space/shell).

$$\Delta q \, \Delta p \geq \frac{\hbar}{2}$$

for 1 particle, volume element

$$(\Delta q_1 \Delta p_1)(\Delta q_2 \Delta p_2)(\Delta q_3 \Delta p_3) \sim h^3 \text{ or } \hbar^3$$

for N particles the volume element would be $\sim h^{3N}$.

If volume of accessible phase space is M,

$$\# \text{ of microstates} = \frac{M}{h^{3N}} = \Omega(E) \quad \text{(finite \& large)}$$

let probability of any microstate $= P$, $\quad \sum_i P_i = 1$

$$P \cdot (\text{no. of microstates}) = 1 \quad \Longrightarrow \quad \begin{cases} P = \dfrac{1}{\Omega} \\ \quad = \dfrac{h^{3N}}{M} \end{cases}$$

Toy model

Set of N coins, where each coin (distinguishable/identifiable) has $\{H, T\}$ two states.

$$\{2^N \text{ microstates}\} \qquad \underbrace{htt \cdots hhh \cdots tth \cdots hhh}_{\text{`N' letters}} \rightarrow \text{ microstate}$$

(still a state of the system as a whole, but describes what each object is doing.)

Let, Total no. of heads $= H$

Total no. of tails $= T$

$$H + T = N$$

Let, $H - T = M$.

(N, M) or (N, H) or (H, T) are macrostate of the system.

For given N,

$$\# \text{ of macrostates} = N + 1.$$

ex for $N = 3$, let (N, H) represent macrostate; we have $(3, 0), (3, 1), (3, 2), (3, 3)$ — 4 macrostates.

no. of microstates are much larger than no. of macrostates.

$$2^N \geq N + 1 \qquad N = \text{d.o.f.}$$

$\rangle$ Probability of any possible/specific microstate would be $\left(\frac{1}{2^N}\right)$.

$\rangle$ Probability of attaining a given macrostate:- we have $0 \le H \le N$

$$P(H) = {}^N C_H \cdot \left(\frac{1}{2}\right)^H \left(\frac{1}{2}\right)^{(N-H)}$$

Suppose the coin is biased, then

$$P(h) = p, \qquad P(t) = 1 - p = q$$

then,

$$P(H) = {}^N C_H \, p^H q^{N-H}$$

$$\sum_{H=0}^{N} P(H) = (p+q)^N = 1^N = 1 \qquad \text{binomial expansion.}$$

Generating function for the binomial distribution

$$f(x) = \sum_{H=0}^{N} P(H)\, x^H = (px+q)^N = \sum_{H=0}^{N} \left({}^N C_H \, p^H q^{N-H}\right) x^H.$$

$P(H) =$ coefficient of x^H in the "expanded" generating function.

Property of G.F.:-

$$f(1) = 1\,; \quad (p+q)^N = 1$$

$$\langle H \rangle = ? \quad \text{(avg. no. of Heads)}$$

$$\langle H \rangle = \frac{\sum_{H=0}^{N} P(H) H}{\sum_{H=0}^{N} P(H)} = \sum_{H=0}^{N} P(H) H$$

to find $\langle H \rangle = \sum_{H=0}^{N} P(H) H$ we differentiate the G.F.

$$f'(x) = \sum_{H=0}^{N} P(H)\, H\, x^{H-1}$$

$$f'(1) = \sum_{H=0}^{N} P(H) H = \Big(N(px+q)^{N-1} \cdot p\Big)_{x=1}$$

so,

$$\langle H \rangle = \sum_{H=0}^{N} P(H) H = N(p+q)^{N-1} \cdot p = Np$$

$$\boxed{\langle H \rangle = Np}$$

$$\langle H^2 \rangle = ?$$

find

$$f''(1) = \Big(Np^2(N-1)(px+q)^{N-2}\Big)_{x=1} = p^2N(N-1) \tag{1}$$

$$f''(x) = \left(\sum_{H=0}^{N} P(H)\,H(H-1)\,x^{H-2}\right)_{x=1} = \langle H(H-1)\rangle = \langle H^2 - H\rangle = \langle H^2\rangle - \langle H\rangle \tag{2}$$

equating (1) and (2)

$$\begin{aligned} p^2N(N-1) &= \langle H^2\rangle - \langle H\rangle \\ \implies \langle H^2\rangle &= p^2N^2 - p^2N + \langle H\rangle \\ \implies \langle H^2\rangle &= p^2N^2 - p^2N + Np \end{aligned}$$

Also, now we can find variance,

$$\text{variance} := \text{mean square} - \text{square of mean}$$

$$\begin{aligned} \mathrm{var}(H) = \langle H^2\rangle - \langle H\rangle^2 \quad &\big(= \langle (H - \langle H\rangle)^2\rangle\big) \\ = N^2p^2 - p^2N + Np - N^2p^2 \\ \mathrm{var}(H) = Np(1-p) \\ \sigma^2 = \mathrm{var}(H) = Npq, \qquad &\text{as } (q = 1-p) \end{aligned}$$

Therefore ΔH = std. deviation = uncertainty = $\sqrt{Npq}$

$$\boxed{\Delta H = \sigma = \sqrt{Npq}}$$

Therefore relative fluctuation will be:-

$$\text{fluctuation about mean} = \frac{\Delta H}{\langle H\rangle} = \frac{\sqrt{Npq}}{Np} = \sqrt{\frac{q}{Np}}$$

$$\text{Relative scatter (dispersion about mean)} = \frac{1}{\sqrt{N}}\sqrt{\frac{q}{p}}$$

$\rangle$ It says relative fluctuation becomes smaller as no. of coin tosses increases / no. of coins increases.

In thermodynamics we deal with $\sim 10^{23}$ d.o.f, so relative fluctuations become very small. This is the main reason "why thermodynamics works".

"In statistical mechanics the generating function is called the partition function ($\sim$ laplace transformation / green fn.)"

Note:- out of 100 coin tosses, $P(50H)$ or $P(50T)$ is most probable, because ${}^{100}C_{50}$ is biggest for all H in $\left({}^{100}C_H\right)$.

each microstate is equally probable, but $P(1H\,\&\,99T)$ has contribution — but each macrostate is made up of many microstates. So attaining $1H$ and $99T$ has a lesser no. of microstates than that of $50H, 50T$; so one example is that one molecule can have nearly all the total energy and rest of all are nearly at rest; indeed such microstates are equally probable, but the macrostate has contribution from very few microstates, so such a macrostate is less likely to exist than other possible macrostates.

2 Lecture 21

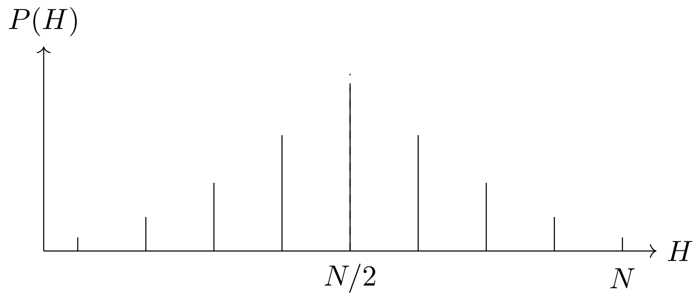
N coins, H heads, T tails

$$P(H) = {}^N C_H \, p^H q^{N-H} \qquad p = \text{prob of heads}, \quad q = \text{prob of tails}$$

$$f(x) = (px+q)^N \qquad\qquad f(x) = \sum_{n=0}^{N} P(n)\, x^n = \sum {}^N C_n \, p^n q^{N-n} x^n = (px+q)^N$$

$$\langle H \rangle = Np \qquad\qquad \Delta H = \sqrt{Npq}$$

What does this distribution look like?



$$P(H) = {}^N C_H \, p^H q^{N-H}$$

we would like to find out what does ${}^N C_H$ do, at large no. of N.

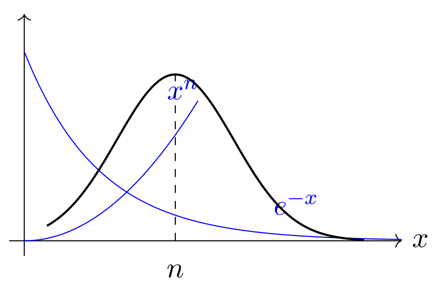
What is $N!$ (for large N):-

Sterling formula:-

$$\begin{aligned} N! &= N(N-1)(N-2)\cdots 3.2.1 \\ &= N^N \left(1 - \frac{1}{N}\right)\left(1 - \frac{2}{N}\right)\cdots \\ &= N^N e^{-N} \sqrt{2\pi N} \cdot \left\{1 + \frac{1}{12N} + \cdots\right\} \end{aligned}$$

$$1! = 1 \cdot e^{-1} \sqrt{2\pi} \qquad \text{so is } e \simeq \sqrt{2\pi}?$$

def:- $n!$ is defined as $\displaystyle\int_0^\infty dx\, e^{-x} x^n = \Gamma(n+1)$ [?] $(n = 1, 2, 3, \dots)$



$$g(x) = e^{-x}x^n = e^{-x}e^{\ln x^n} = e^{-x+n\log_e x} = e^{-(x-n\ln x)}$$

$$g'(x) = e^{-(x-n\ln x)} \cdot (-1)\left(1 - \frac{n}{x}\right) = 0 \implies \boxed{g'(n) = 0}$$

let, $f(x) = x - n\ln x$

$$f(n) = n - n\ln n$$

$$g'(x) = 1 - \frac{n}{x}, \qquad g''(x) = 0 + \frac{n}{x^2}$$

$$f(x) = f(n) + \frac{f'(n)(x-n)}{1!} + \frac{f''(n)(x-n)^2}{2!} + \cdots$$

$$f(x) = (n - n\ln n) + 0 + \frac{(x-n)^2}{2n} + \cdots$$

$$n! = \int_0^\infty e^{-x}x^n\,dx = \int_0^\infty e^{-(x-n\ln x)}\,dx$$

$$\implies n! = \int_0^\infty e^{-\left((n-n\ln n)+\frac{(x-n)^2}{2n}+\cdots\right)}dx$$

$$= \int_0^\infty \left(e^{-n}n^n\right)e^{-\frac{(x-n)^2}{2n}}\left\{1+\cdots\right\}dx$$

for very large n

$$n! = e^{-n}n^n\int_{-\infty}^{\infty}dx\,e^{-\frac{(x-n)^2}{2n}} \qquad \left(\int_{-\infty}^{\infty}e^{-t^2/A}\,dt = \sqrt{\pi A}\right)$$

$$= e^{-n}n^n\sqrt{2\pi n}\cdot\left\{1 + O\left(\frac{1}{n}\right)\right\}$$

Gaussian integral

$$I = \int_{-\infty}^{\infty}e^{-ax^2}\,dx = \sqrt{\pi/a}$$

also

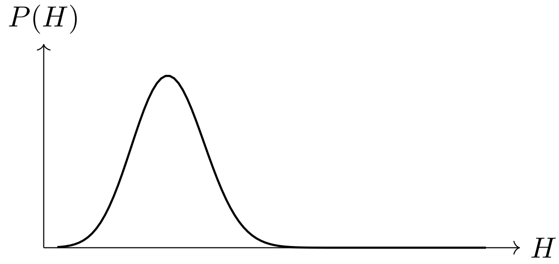
$$I = \int_{-\infty}^{\infty}e^{-ay^2}\,dy$$

$$I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}e^{-a(x^2+y^2)}\,dx\,dy \qquad \begin{cases} x^2+y^2 = r^2 \\ \qquad x = r\cos\theta\,, \quad dA = (r\,d\theta)\,dr \\ \qquad y = r\sin\theta \end{cases}$$

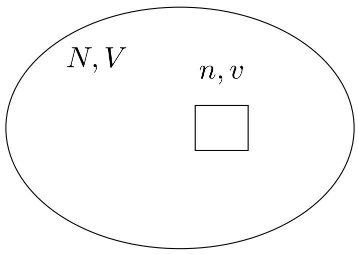
$$= \int_0^\infty\int_0^{2\pi}e^{-ar^2}\,r\,dr\,d\theta = \int_0^\infty e^{-ar^2}\,r\,dr\int_0^{2\pi}d\theta = \left(\sqrt{\pi/a}\right)^2$$

$$\left(I = \sqrt{\pi/a}\right)$$

So for large N:



Take an ideal gas contained in volume V. N = total no. of particles.



What is the probability that v contains n particles?

$P(n)$ = Probability that v contains n particles.

Probability that a given particle is in volume $v = v/V$.

Assume that all the particles are moving independent of each other; therefore the probability that there are n of them inside v is

$$= {}^N C_n \left(\frac{v}{V}\right)^n \left(1 - \frac{v}{V}\right)^{N-n}$$

Here n is a random variable whose value can vary from $(0 \le n \le N)$.

here $p = v/V, \qquad q = 1 - v/V.$

this is the binomial distribution.

let's put $\rho = N/V$

$$\implies \frac{1}{V} = \frac{\rho}{N}$$

so,

$$P(n) = {}^N C_n \left(\frac{v\rho}{N}\right)^n \left(1 - \frac{v\rho}{N}\right)^{N-n} \qquad \text{(binomial distribution)}$$

What happens to $P(n)$, when $N \to \infty$ and $V \to \infty$ but keeping ρ fixed.

$$P(n) \xrightarrow{N\to\infty} \frac{N!}{n!(N-n)!} \left(\frac{v\rho}{N}\right)^n \left(1 - \frac{v\rho}{N}\right)^{N-n}$$

$$= \frac{N^N e^{-N}\sqrt{2\pi N}}{\left(n^n e^{-n}\sqrt{2\pi n}\right)\left((N-n)^{N-n} e^{-(N-n)}\sqrt{2\pi(N-n)}\right)} \left(\frac{v\rho}{N}\right)^n \left(1 - \frac{v\rho}{N}\right)^{N-n}$$

(we can't use $n! = e^{-n} n^n \sqrt{2\pi n}$ as n can take 0 too)

$$P(n) \longrightarrow \frac{e^{-\bar{n}}(\bar{n})^n}{n!} \qquad \text{where } n = 0, 1, 2, 3, \ldots \infty$$

$$\boxed{v\rho = \bar{n}}$$

and $\bar{n} = \rho v$, = avg. no. of particles in sub-volume v.

$$(\bar{n} = Np = (N \cdot v/V)) = (\bar{n} = \rho v)$$

We have assumed that particles behave like Newtonian mech. particles and are distinguishable from one another.

In $\left(P(n) = \dfrac{e^{-\rho v}(\rho v)^n}{n!}\right)$ formula, (N, V) don't appear anymore, as $(N, V \to \infty)$ such that their ratio $\rho = N/V$ is finite (this is called the thermodynamic limit of the system, when $N \to \infty, V \to \infty$ such that $\rho = N/V$ is finite). Statistical mech. reduces to thermodynamics in the thermodynamic limit.

Poisson Distribution

for random variable $n = 0, 1, 2, \ldots \infty$

$$P(n) = \frac{e^{-\lambda}\lambda^n}{n!} \qquad (\lambda = \langle n \rangle) \quad \lambda \text{ is avg. value of random variable}$$

Generating function

$$f(x) = \sum_{n \geq 0} P(n) x^n = e^{\lambda(x-1)} \qquad \text{(How?)}$$

$$f(1) = 1 \qquad \text{for} \qquad \sum_{n \geq 0} P(n) = 1$$

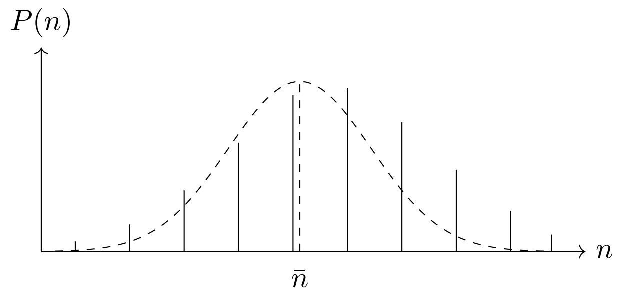
$$\text{variance} = \langle (n - \langle n \rangle)^2 \rangle = \lambda = \langle n \rangle = \langle n^2 \rangle - \langle n \rangle^2$$

In poisson distribution, variance = mean.

$$\text{Relative fluctuation} = \frac{\sigma}{\langle n \rangle} = \frac{\Delta n}{\langle n \rangle} = \frac{\sqrt{\langle n \rangle}}{\langle n \rangle} = \frac{1}{\sqrt{\langle n \rangle}}$$

for poisson's distribution, $\langle n \rangle = \lambda$, $(\Delta n)^2 = \lambda$, & all higher moments $= \lambda$, so it's a single-parameter distribution;

In gaussian distribution, except for $\langle n \rangle$ & $(\Delta n)^2$, all higher moments vanish.



$\bar{n}$ = avg. no. of particles in subvolume v.

If $\bar{n}$ itself becomes very large, then the deviation from mean $(n - \bar{n})$ takes a gaussian curve, and we call that a continuous variable if $\bar{n}$ is very large.

from $P(n) = \dfrac{e^{-\bar{n}}(\bar{n})^n}{n!}$

using Sterling's formula again, and replacing the variable to $x = n - \bar{n}$

$$P(x) \propto \frac{e^{-x^2/2\sigma^2}}{\sqrt{2\pi}\,\sigma} \qquad \text{(Gaussian distribution)}$$

"The whole point of the above discussion is that if we start with a binomial distribution of Bernoulli trials, and then from that we take the no. of particles (trials) to be very large such that $p(H) \to 0$ (probability of success in a trial is vanishingly small), then the Binomial distribution goes over into the Poisson distribution. Then if the mean value of the Poisson distribution is very large compared to unity, the deviation from the mean $(n - \bar{n})$ is approximately a continuous variable that has a Gaussian shape."

So a probability distribution can shift over to another probability distribution.

Generating function:-

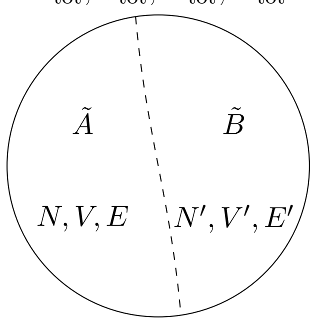
$$f(x) = \sum_{n \geq 0} P(n)x^n = P(0)x^0 + P(1)x^1 + P(2)x^2 + \cdots + P(n)x^n$$

$$f(x) = \sum_{n \geq 0} P(n)x^n = \sum_{n \geq 0} \frac{e^{-\lambda}\lambda^n}{n!} x^n \implies e^{-\lambda} \sum \frac{(\lambda x)^n}{n!} = e^{-\lambda} e^{\lambda x}$$

$$\boxed{f(x) = e^{\lambda(x-1)}}$$

let's go back to our problem of an isolated system in thermal equilibrium.

$N_{\text{tot}},\ V_{\text{tot}},\ E_{\text{tot}},\ \Omega_{\text{tot}}$



no. of accessible microstate.

Imagine the system is made of two sub-systems, in eq. thermal eq^m with each other, such that

$$N + N' = N_{\text{tot}}, \qquad V + V' = V_{\text{tot}}, \qquad E + E' \cong E_{\text{tot}} \qquad \text{(Isolated, in thermal eq}^m\text{)}$$

(*not necessarily*: $E + E' \neq E_{\text{tot}}$ always if pot. energy plays a role.)

$E + E' \cong E_{\text{tot}}$ to a very good approximation.

"as no. of d.o.f which are interacting near the partition is $\left(\dfrac{1}{10^8}\right)$ *th of the total d.o.f"* (assuming short range forces).

Then at any instant of time, what is the probability that Energy of A is E; $P(E)$

$P(E) =$ no. of microstates of this entire system such that A has energy E. $(\Omega(E))$ (also B has energy E'.)

$$= \frac{\Omega(E)\,\Omega'(E')}{\Omega_{\text{tot}}(E_{\text{tot}})} \qquad \text{(normalization)}$$

$\Omega(E)$ & $\Omega'(E')$ can have totally different functions [?].

As A can be a jar of a large container of oil and B can be our atmosphere. So A, B have different d.o.f.

3 Lecture 22

Note that Energy of A and B are not fixed. Because there are fluctuations.

E, V, N, N', V', E' are variable quantities.

What we were trying to do is to find, is

"Given the postulate of equal a priori probabilities of all the accessible microstates of the total system, what can we say about the probability distribution of (for instance say) Energy in system A."

$$P(E) = \frac{\#\text{ of microstates of tot. system such that } A \text{ has energy } E \text{ (\& } B \text{ has } E')}{\text{Total no. of microstates of total system such that } T.E = E_{\text{tot}}.}$$

If A & B are assumed as independent systems (they actually are not) then $\Omega_{\text{tot}}(E) = \Omega(E) \cdot \Omega'(E')$

$$= \frac{\Omega(E) \cdot \Omega'(E')}{\Omega_{\text{tot}}(E_{\text{tot}})}.$$

$$P(A \,\&\, B) = P(A) \cdot P(B) \quad \text{only iff } A, B \text{ are independent events.}$$

In the above discussion we have assumed that the interaction Energy is fairly negligible.

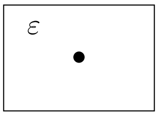
$\Omega(E)$ is actually a function of V, N also.

<u>Remember always:</u>

$$\begin{cases} \Omega(E) = \Omega(E, V, N) \\ \Omega'(E') = \Omega'(E', V', N') \end{cases}$$

Let's see how big these numbers are:

Suppose we take a single particle and put it into volume V, and the particle is moving around with some energy $\leq E$. Let's calculate how many microstates it has.



V

No. of microstates is the no. of shells in phase-space (true in any dimensions)

$$\underbrace{E = \frac{p^2}{2m}} \qquad |p| \leq \sqrt{2mE}$$

Let no. of such microstates be

$$\phi(E) = \frac{1}{h^3} \int d^3q \int_{|p|\leq\sqrt{2mE}} d^3p$$

where $d^3p = 4\pi p^2\, dp$ is the volume element, and $\dfrac{\text{phase space volume}}{\text{min. volume } (h^3)}$ gives the number of microstates.

$$\begin{aligned}
\phi(E) &= \frac{V}{h^3}\, 4\pi \int_0^{\sqrt{2mE}} p^2\, dp \;\propto\; \frac{V}{h^3} \qquad \{\text{no. of dimensions}\} \\
&= \frac{V}{h^3}\, \frac{4\pi}{3}\, p^3\Big|_0^{\sqrt{2mE}} \\
&= \frac{V}{h^3}\, \frac{4\pi}{3}\, \varepsilon^{3/2}\, 2m^{3/2} \qquad (p \propto E^{1/2})
\end{aligned} \tag{1}$$

Relation between E and p for a free particle is

$$E^2 = c^2p^2 + m^2c^4$$

$m =$ rest mass always.

Mass does not increase with speed, it's just $p = \gamma m v$ instead of mv. $\left(\gamma = \dfrac{1}{\sqrt{1 - v^2/c^2}}\right)$

$$\varepsilon = (c^2p^2 + m^2c^4)^{1/2} = c(p^2 + m^2c^2)^{1/2}$$

If $p \ll mc$ $\qquad\qquad \{mc = \text{Compton momentum}\}$

$$\varepsilon = mc^2 \left(1 + \frac{p^2}{m^2c^2}\right)^{1/2}$$

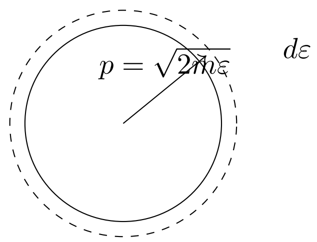
In n dimensions, $(n-1)$ angles vary for $(0 - \pi$ or $0 - 2\pi)$.

In n-spatial dimensions,

$$\phi(\varepsilon) \propto \int_0^{\sqrt{2m\varepsilon}} p^{n-1}\, dp \;\propto\; \varepsilon^{n/2}$$

$$\begin{aligned}
\varepsilon &= mc^2 \left(1 + \frac{p^2}{2m^2c^2} - \frac{1}{8}\frac{p^4}{m^4c^4} + \cdots\right) \\
&= mc^2 + \frac{p^2}{2m} + \cdots = \text{rest mass energy} + \frac{p^2}{2m}
\end{aligned} \tag{2}$$

No. of microstates with energy in $(\varepsilon, \varepsilon + d\varepsilon) \sim \varepsilon^{1/2} d\varepsilon = d\phi(\varepsilon)$



(given energy shell)

$$d\Omega = 4\pi p^2\, dp \qquad d\nu \simeq 8\pi\, m^{3/2} \varepsilon^{1/2}\, d\varepsilon$$

$$\omega(\varepsilon) \sim \varepsilon^{1/2} d\varepsilon = f(\varepsilon)\, d\varepsilon$$

Density of states $f(\varepsilon) = \varepsilon^{1/2} = g_i$ (degeneracy at energy ε) {no. of microstates per unit energy interval at energy ε}

For n-D: $\Omega \propto \int p^{n-1} dp$ $(\Omega \propto \varepsilon^{n/2})$ for $(n \geq 2)$.

Case for 2-D container

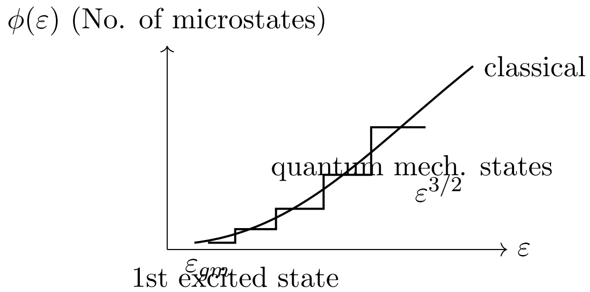
$$d\nu = 2\pi p\, dp = 2\pi \sqrt{2m\varepsilon}\, \frac{2m}{2\sqrt{\varepsilon}} = \text{constant}$$

density of states becomes constant $\propto$ 2-D.

Case for 1-D

$$\phi(\varepsilon) = \frac{d}{d\varepsilon}\left(\frac{V}{h^3}\varepsilon^{1/2}\right) = \frac{1}{2}\varepsilon^{-1/2}\frac{V}{h^3}$$

Density of states $\uparrow$ as $\varepsilon \downarrow$



As $\varepsilon \uparrow$ no. of states increases [3-D case] (microscopic)

Now, we want to find $\Omega(E)$, so we will like to assume again that particles are free and

$$E_{\text{tot}} = \varepsilon_1 + \varepsilon_2 + \varepsilon_3 + \cdots + \varepsilon_N$$

Sum of energy of individual particles (with index i) = total energy.

For any two particles having z states, together they occupy $z^2 = q$ states.

$$\Omega(E) \propto \varepsilon \propto E^{\alpha N}$$

If one particle can take $\phi(\varepsilon)$ states with energy ε to $\varepsilon + d\varepsilon$: $= k\varepsilon^{3/2}$; independent N particles will take $- \; k\varepsilon^{(3/2)N}$.

If all ε_i are of same order of magnitude ε {approximation}

$$E = N\varepsilon$$

So,

$$\Omega(E) \propto \varepsilon^{\alpha N} \propto \left(\frac{E}{N}\right)^{\alpha N}$$

For very large N, $\Omega(E)$ is astronomically large number.

$\alpha = \frac{3}{2}\, n\, \frac{1}{2}\, n\, \frac{5}{2} \cdots$

lecture 22 (time 30:15)

4 Lecture 22

Note: For Most probable "macro" state i.e. $P(E)$ is maximum $\equiv$ {eqm state}

So the system which has largest no. of microstates, that macrostate is the most probable state.

For Most probable macrostate it implies:

$$\frac{\partial P(E)}{\partial E} = 0 \qquad \left\{ P(E) = \frac{\Omega(E)\,\Omega'(E')}{\Omega_{\text{tot}}(E_{\text{tot}})} \right\}$$

As $P(E)$ is very large, let's take $\log P(E)$ and differentiate.

$$\frac{\partial}{\partial E}\big(\ln P(E)\big) = 0$$

$$dP = \frac{\partial P}{\partial E} dE + \frac{\partial P}{\partial V} dV + \frac{\partial P}{\partial N} dN \qquad \frac{\partial(\ln \Omega(E))}{\partial E} + \frac{\partial(\ln \Omega'(E'))}{\partial E} = 0 \quad \Bigg\}$$

for eqm $\left(\dfrac{\partial P(E)}{\partial E} = \dfrac{\partial P}{\partial V} = \dfrac{\partial P}{\partial N} = 0\right)$

$$E + E' = E_{\text{tot}} \qquad E' = E_{\text{tot}} - E \;\Rightarrow\; dE' = -dE$$

therefore

$$\boxed{\frac{\partial \ln(\Omega(E))}{\partial E} = \frac{\partial \ln \Omega'(E')}{\partial E'}}$$

cond. for thermal eqm./most probable macrostate.

$A \,|\, B$ possible, among two different kinds of matter, fluid, (gas $\to$ liquid)...

$$\frac{\partial}{\partial E}\big(\ln \Omega(E)\big) = \frac{\partial}{\partial E'}\big(\ln \Omega'(E')\big) = [E^{-1}] \quad \text{(dimensions)}$$

or,

$$\text{some property of A} = \text{some property of B} = \frac{1}{k_B\,\text{Temp}} \qquad \text{\{Imp.\}}$$

defn of temperature:

$$(dE = TdS - PdV + \mu dN) \qquad T = \left.\frac{\partial E}{\partial S}\right|_{V,N} \quad , \quad S = k_B \ln \Omega$$

Let's define temp so for this property, a function $\beta(E)$, of A/B.

$$\boxed{\beta(E) = \frac{\partial \ln \Omega(E)}{\partial E} = \frac{1}{k_B\, T(E)}}$$

By defn, in equilibrium

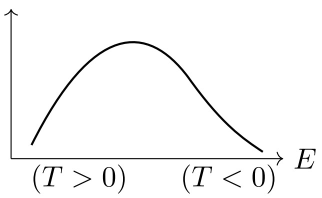
$$\boxed{T_A = T_B}$$

"Any two parts have same temp in large isolated system under thermal equilibrium." (large d.o.f)

"Of course each part must be large enough to talk about temp, as temp. is not defined for a single particle."

If $\Omega(E)$ is not strictly increasing function, and if so

$$\frac{\partial \ln(\Omega(E))}{\partial E} < 0 \;\Rightarrow\; \text{negative temperature.}$$



(+ve slope implies positive temp; -ve slope implies negative temp.) So possibility of $(-\text{ve})$ temp is built in our defn. of temp.

Since $\Omega = \Omega(N, V, E)$, from $\left(\frac{\partial \ln \Omega}{\partial V} = 0,\ \frac{\partial \ln \Omega}{\partial N} = 0\right)$

$$\frac{\partial \ln \Omega}{\partial V} = \frac{\partial \ln \Omega'}{\partial V'} \equiv \left\{\left(\frac{P_A}{T_A} = \frac{P_B}{T_B}\right) \Rightarrow (P_A = P_B)\right\}$$

$$\frac{\partial \ln \Omega}{\partial N} = \frac{\partial \ln \Omega'}{\partial N'} \equiv \left\{\left(\frac{\mu_A}{T_A} = \frac{\mu_B}{T_B}\right) \Rightarrow (\mu_A = \mu_B)\right\}$$

(in thermodynamic terms)

$$E = TS - PV + \mu N \qquad S = \frac{E + PV - \mu N}{T}$$

$$\left(\frac{\partial S}{\partial N} = -\frac{\mu}{T}\right) \; ; \; \left(\frac{\partial S}{\partial V} = \frac{P}{T}\right)$$

Entropy

$$S(E, V, N) = k_B \ln\big(\Omega(E, V, N)\big)$$

Thermodynamics in Nutshell

1st law of thermodynamics

$$dQ = dU + dW \qquad \{dQ = \text{not a perfect differential}\}$$
$$dU = dQ - dW$$
$$= dU + PdV - \mu\, dN \qquad \text{(generalized fluxes)}$$
$$dQ = dU - \sum F_i\, dx_i \qquad \text{(generalized forces)}$$

(Entropy representation)

$$dS = \frac{dE + PdV - \mu dN}{T} \qquad S = S(E, V, N)$$

(Energy representation)

$$dE = TdS - PdV + \mu dN \;\Rightarrow\; E = E(S, V, N)$$

5 Lecture 23

'E' is some kind of thermodynamic potential

$$E = E(S, V, N) \qquad E = \text{Energy}$$

$$\left.\begin{aligned} T &= \left(\frac{\partial E}{\partial S}\right)_{V,N} \\ P &= -\left(\frac{\partial E}{\partial V}\right)_{S,N} \\ \mu &= \left(\frac{\partial E}{\partial N}\right)_{S,V} \end{aligned}\right\} \Longleftarrow dE = TdS - PdV + \mu dN$$

What is significance of above eqn set?

We know that in $E(S, V, N)$, 'S' is not a control variable. So let's change that by Legendre transformations.

$$E = E(S, V, N) \;\longrightarrow\; \text{(Internal Energy)}$$

<u>Assumption</u>: "E is homogeneous function of thermodynamic variables S, V, N."

For that we have to assume we are in the thermodynamic limit of the thermodynamic system, i.e. $V \to \infty$ and $N \to \infty$ with $N/V = s$ (fixed).

Then S, V, N, which increase with system size (extensive quantities), lead to some E. Then we can assume E is a homogeneous function of S, V, N of degree 1.

Euler's theorem

For any homogeneous function of degree 'r'

$$f = f(x, y, z, \ldots)$$

i.e. $f(\lambda x, \lambda y, \lambda z, \ldots) = \lambda^r f(x, y, z, \ldots)$.

$$\frac{\partial}{\partial \lambda}\left(x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} + z\frac{\partial f}{\partial z} + \cdots\right) = rf$$

Then for $E = E(S, V, N)$:

$$S\left(\frac{\partial E}{\partial S}\right)_{V,N} + V\left(\frac{\partial E}{\partial V}\right)_{S,N} + N\left(\frac{\partial E}{\partial N}\right)_{S,V} = E$$

$$S(T) + V(-P) + N(\mu) = E \qquad \text{(Euler's relation)}$$

So E must be equal to $TS - PV + \mu N$.

$$\text{or} \quad E - TS + PV = \mu N$$

we also know that $\big(G(T, P, N) = E - TS + PV\big)$

so $\mu N = G$

so $\Rightarrow \mu = \dfrac{G}{N}$ = Gibbs free energy per particle.

Let us find dE:

$$dE = (TdS - PdV + \mu dN) + (SdT - VdP + Nd\mu)$$

But laws of thermodynamics tell us that

$$(dE = TdS - PdV + \mu dN) \text{ only}$$

so it implies that

$$SdT - VdP + N\,d\mu = 0$$

$$d\mu = -\frac{V}{N}dP + \frac{S}{N}dT \quad \Longrightarrow \quad d\mu = v\,dP - s\,dT$$

<u>specific volume per particle</u> <u>specific entropy per particle</u>

So,

$$d\mu = v\,dP - s\,dT \qquad \text{(Gibbs–Duhem relation)}$$

it implies,

$$\boxed{\mu = \mu(P, T)}$$

So once we know (P, T) we know the chemical potential (μ).

If we assume 'T' is some kind of generalized flux force and 'dS' as some generalized flux, then

$$dE = T\,dS + \sum F_i\,dX_i$$

$$dE = \sum F_i\,dX_i \qquad \text{(homogeneity argument)}$$

so:

$$\mu N + PV - E = \sum F_i X_i \;\Rightarrow\; \left(\sum X_i\,dF_i = 0\right) \qquad \text{(Generalized Duhem relation)}$$

This confirms (for our assumption of E to be a homogeneous function of degree 1. If E is not homogeneous function, so $\mu = G/N$, degree $\neq 1$.

Legendre transforms

$$E(S,V,N) \longrightarrow H(S,P,N)$$

(Enthalpy)

$$H = E + PV$$

$$\begin{aligned} dH &= dE + PdV + VdP \\ &= (TdS - PdV + \mu dN) + PdV + VdP \\ dH &= TdS + VdP + \mu dN \end{aligned} \tag{3}$$

So $H = H(S,P,N)$.

$$E(S,V,N) \longrightarrow F(T,V,N)$$
$$F = E - TS$$

$$\begin{aligned} dF &= TdS + \mu dN - PdV - TdS - SdT \\ dF &= -SdT - PdV + \mu dN \;\Rightarrow\; F = F(T,V,N) \end{aligned} \tag{4}$$

$$E(S,V,N) \longrightarrow G(T,P,N) \qquad G = E - TS + PV \qquad \text{(Gibbs free energy)}$$

$$E(S,V,N) \longrightarrow \Phi_1(S,V,\mu) \qquad \Phi_1 = E - \mu N$$

$$E(S,V,N) \longrightarrow \Phi_2(S,P,\mu) \qquad \Phi_2 = E + PV - \mu N$$

$$E(S,V,N) \longrightarrow \Phi_3(T,V,\mu) \qquad \Phi_3 = E - TS - \mu N$$

But we know $E = TS - PV + \mu N$, so $E - TS - \mu N = -PV$.

So

$$\boxed{\Phi_3 = -PV} \;\rightarrow\; \text{“grand potential”}$$

The reason is that this potential Φ_3 is connected to the grand canonical ensemble, but the other potentials above mentioned are only linked to the canonical ensemble of stat. mech.

In,

$$E = TS - PV + \mu N$$

$(T,P,\mu) \rightarrow$ Intensive variables; $(S,V,N) \rightarrow$ Extensive variables.

Field variables (Intensive)	**State variables (extensive)**
T	S
P	V
μ	N
Stress σ_{ij}	Strain ϵ_{ij}
$\vec{E}$	$\vec{P} \rightarrow$ polarisation per unit volume
$\vec{H}$	$\vec{M} \rightarrow$ magnetic dipole moment per unit volume

$$dF = -SdT - PdV + \mu dN \ ; \qquad dG = -SdT + VdP + \mu dN$$

Let's look at these 2 relations,

$$S = -\left(\frac{\partial F}{\partial T}\right)_{V,N}$$

So,

$$\left(\frac{\partial F}{\partial T}\right)_{V,N} = \left(\frac{\partial G}{\partial T}\right)_{P,N}$$

Similarly,

$$P = -\left(\frac{\partial F}{\partial V}\right)_{T,N} \qquad \mu = \left(\frac{\partial F}{\partial N}\right)_{T,V}$$

It says these thermodynamic potentials $(U, G, F, H\ldots)$, their 1st partial derivative with their individual variables on which they depend, are other thermodynamic variables.

$$dQ = dU + dW \qquad TdS = dE + PdV - \mu dN$$

$$\left(\frac{dQ}{dT}\right)_{V,N} = C_V = T\left(\frac{\partial S}{\partial T}\right)_{V,N} = \left(\frac{\partial E}{\partial T}\right)_{V,N} + 0 - 0$$

$$C_V = T\left(\frac{\partial S}{\partial T}\right)_{V,N} \quad , \qquad S = -\left(\frac{\partial F}{\partial T}\right)_{V,N}$$

So

$$\boxed{C_V = -T\left(\frac{\partial^2 F}{\partial T^2}\right)_{V,N}}$$

the response functions are 2nd derivatives of thermodynamic potentials.

But we know C_V cannot be negative $\rightarrow$ Le Chatelier's principle.

So,

$$T\left(\frac{\partial^2 F}{\partial T^2}\right) < 0.$$

$$C_P = T\left(\frac{\partial S}{\partial T}\right)_{P,N} = -T\left(\frac{\partial^2 G}{\partial T^2}\right)_{P,N} \qquad \text{as } S = -\left(\frac{\partial G}{\partial T}\right)_{P,N}$$

also

$$(C_P > C_V) \ \Rightarrow \ \left|\frac{\partial^2 G}{\partial T^2}\right| > \left|\frac{\partial^2 F}{\partial T^2}\right|.$$

Isothermal Compressibility (fixed T, N)

$$K_T = \frac{1}{\text{Bulk modulus}} \qquad (\text{as } P\uparrow,\ V\downarrow)$$

$$K_T = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_{T,N}$$

Since $\left(V = \left(\frac{\partial G}{\partial P}\right)_{T,N}\right)$,

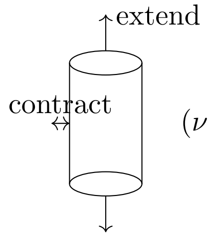
$$\left(\frac{\partial V}{\partial P}\right)_{T,N} = \left(\frac{\partial^2 G}{\partial P^2}\right)_{T,N}$$

$$\boxed{K_T = -\frac{1}{V}\left(\frac{\partial^2 G}{\partial P^2}\right)_{T,N}}$$

Poisson's ratio (ν)

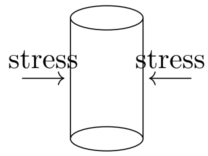
Poisson's ratio is a measure of the Poisson effect, that describes the expansion or contraction of a material in the direction perpendicular to the direction of loading.

$$\nu = \frac{\Delta x}{\Delta y} = \frac{\text{contraction strain}}{\text{extension strain}}$$



$(\nu > 0)$

It is possible that ($\nu < 0$) such that upon extending in one direction, the object/material also expands in the $\perp$ direction.



extending in $\perp$ direction

It turns out

$$\boxed{-1 < \nu < \tfrac{1}{2}}$$

For ideal gas

$$PV = nRT \qquad PdV + VdP = nRdT$$

at constant T: $d(PV) = -PdV$, i.e. $VdP = -PdV$

$$\frac{dV}{V} = -\frac{dP}{P} \;\Rightarrow\; K_T = -\frac{1}{V}\frac{\partial V}{\partial P} = -\frac{1}{V}\left(\frac{-V}{P}\right) = \frac{1}{P}$$

If K_T is high, pressure is low.

For adiabatic process

$$PV^\gamma = \text{const.}$$

$$dP\,V^\gamma + P\gamma V^{\gamma-1}dV = 0$$

$$V^{\gamma-1}\left(VdP + \gamma PdV\right) = 0$$

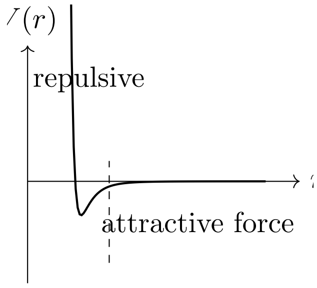
$$\Rightarrow\; K_T = -\frac{1}{V}\frac{dV}{dP} = \frac{1}{\gamma}\cdot\frac{V}{VP} = \frac{1}{\gamma P}$$

$$\left(K_{S,N} = \frac{1}{\gamma P}\right)$$

Van der Waals gas

What happens if we have a van der Waals gas:

$$P = \frac{nRT}{V - bn} - \frac{an^2}{V^2}$$



This repulsion arose from the Pauli exclusion principle ($r > 0$).

Attractive force potential: "6–12" potential (Lennard-Jones potential)

$$V(r) = V_0 \left[\left(\frac{a}{r}\right)^{12} - \left(\frac{a}{r}\right)^{6} \right] \qquad \text{need quantum mech. effect (repulsive term).}$$

Real gas approximates to van der Waals gas.

Attractive force term arises from dipole moment attraction.

Random fluctuations can cause dipole moment.

Interaction energy of any dipole goes like

$$U(r) \sim \vec{p} \cdot \vec{E}$$

the greater the field, the greater the separation.

So $\vec{p} = \alpha \vec{E}$

$$U \sim \alpha |E|^2 \qquad \vec{E} \text{ of dipole } \sim \frac{2kp}{r^3}$$

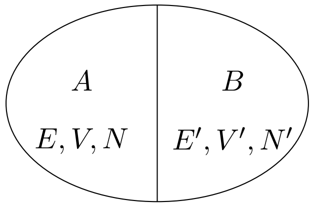
$$U \propto E^2 \propto \frac{1}{r^6}$$

This formula holds true only for molecules with temporary dipole moment.

6 Lecture 24

Let us try to understand the distribution of energy in a subsystem of a (large isolated system).

Isolated system $(E_{\text{tot}}, V_{\text{tot}}, N_{\text{tot}})$



cond. of eqm implies,
$$\frac{\partial \ln \Omega}{\partial E} = \frac{\partial \ln \Omega'}{\partial E'} = \frac{1}{k_B T} \qquad (T_A = T_B)$$

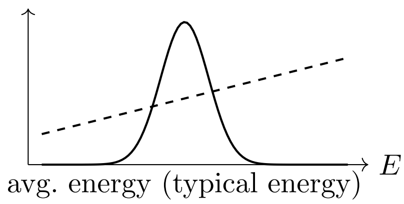
Probability that subsystem A has energy E:

$$P(E) = \frac{\text{No. of microstates of total system such that } A \text{ has energy } E}{\text{Total no. of accessible microstates of system with energy } E_{\text{tot}}}$$

$$P(E) = \frac{\Omega(E) \cdot \Omega'(E')}{\Omega_{\text{tot}}(E_{\text{tot}})}$$

where $(0 \leq E \leq E_{\text{tot}})$.

We know that
$$\Omega(E)\, \Omega'(E') \propto \Omega'(E_{\text{tot}} - E) \propto (E_{\text{tot}} - E)^N$$



$$\ln P(E) = \ln \Omega(E) + \ln \Omega'(E_{\text{tot}} - E) - \ln \Omega_{\text{tot}}(E_{\text{tot}})$$

Let us expand $\ln P(E)$ about some point $\bar{E}$ or $\langle E \rangle$.

$$\ln P(E) \approx \ln P(\bar{E}) + (E - \bar{E})\, \frac{\partial \ln P(E)}{\partial E}\bigg|_{E=\bar{E}} + \frac{(E-\bar{E})^2}{2!}\, \frac{\partial^2 \ln P(E)}{\partial E^2}\bigg|_{E=\bar{E}}$$

Since $P(E)$ is max at $E = \bar{E}$, $\displaystyle \frac{\partial \ln P(E)}{\partial E}\bigg|_{\bar{E}} = 0.$

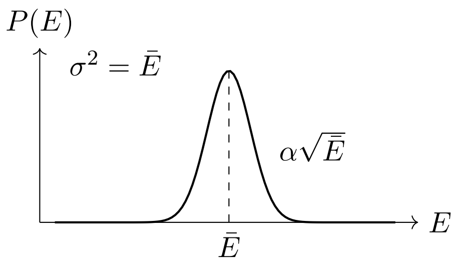
$$\ln P(E) = \ln P(\bar{E}) + \frac{(E-\bar{E})^2}{2!}\, \frac{\partial^2 \ln P(E)}{\partial E^2}\bigg|_{\bar{E}}$$

Since the slope of the slope at $\bar{E}$ is negative,

$$= -k\, \frac{(E-\bar{E})^2}{2}$$

$$P(E) = P(\bar{E}) \cdot e^{-k(E-\bar{E})^2/2} \qquad \text{‘}k\text{’ to make dimensionless}$$

$$P(E) = (\text{Normalisation const.}) \cdot e^{-(E-\bar{E})^2/(2\bar{E})}$$



So relative fluctuations go like $\dfrac{\sqrt{\bar{E}}}{\bar{E}} = \dfrac{1}{\sqrt{\bar{E}}}$.

$$\left(\bar{E} = N\,\frac{f}{2}k_B T\right)$$

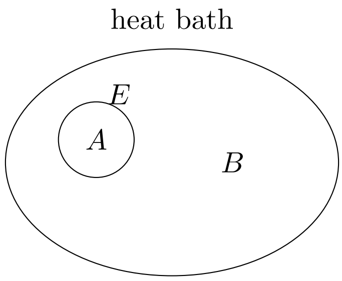
relative fluctuations about mean $\propto \dfrac{1}{\sqrt{N}}$

So for $N \sim 10^{24}$, fluctuations in eqm about mean value in avg. energy $\propto 10^{-12}$, which is negligible. That is why in thermodynamics we only talk about $\bar{E}$ or $\langle E \rangle$ rather than E_{tot}.

So internal energy used in thermodynamics is actually the average energy, but when we study statistical mech., we have to become precise, and talk about E and $\bar{E}$.

(B) What if 'A' is a very small ensemble/subsystem sitting in a huge heat bath (B)?

'A' might contain a few atoms or a single atom/molecule, so energy of 'A' will have very large fluctuations from its mean value $\langle E \rangle$.



Let us assume total energy of A is E, and that of B is E': $(E + E' = E_{\text{tot}})$, $(E' = E_{\text{tot}} - E)$.

$$\begin{aligned} P(E) &= \frac{\text{No. of microstates of total system such that } A \text{ has energy } E}{\text{Total no. of accessible microstates of total system with energy } E_{\text{tot}}} \\ &\neq \frac{\text{No. of microstates of } A \text{ with energy } E \text{ \& No. of microstates of } B \text{ with energy } E'}{\text{Total } \Omega_{\text{tot}}(E_{\text{tot}})} \end{aligned}$$

Since $(E \ll E_{\text{tot}})$, A is strongly affected by B, but B is not affected by A at all (as $N_A \ll N_B$).

So,

$$P(E) = \frac{\Omega'(E')}{\Omega_{\text{tot}}(E_{\text{tot}})}.$$

$$\ln P(E) = \ln \Omega'(E') - \ln \Omega_{\text{tot}}(E_{\text{tot}}) = \ln \Omega'(E_{\text{tot}} - E) - \ln \Omega_{\text{tot}}(E_{\text{tot}})$$

Since ($E_{\text{tot}} \gg E$), let us expand $\ln \Omega'(E')$ about (E_{tot}).

$$\ln P(E) = \left[\ln \Omega'(E_{\text{tot}}) - E \underbrace{\left.\frac{\partial \ln \Omega'(E')}{\partial E'}\right|_{E'=E_{\text{tot}}}}_{\beta} + \cdots \right] - \ln \Omega_{\text{tot}}(E_{\text{tot}})$$

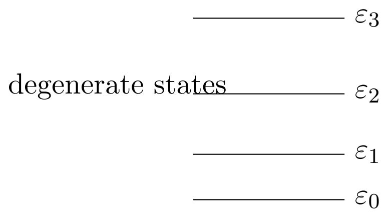
$$P(E) = (\text{const})\, e^{-\beta E} \qquad \beta = \frac{\partial \ln \Omega'(E')}{\partial E'} = \text{inverse temp} = \frac{1}{k_B T} \text{ of heat bath}$$

To find (const) we need to know $\Omega'(E_{\text{tot}})$ (info about heat bath). But we don't need that either — we can normalize $P(E)$ and find the normalisation const.

$$P(E) = \frac{e^{-\beta E}}{\sum_E e^{-\beta E}}$$

So, normalisation const. is fixed by properties of the subsystem alone, and we eliminate whatever happens outside.

For a moment, let us assume 'A' has discrete energy levels.

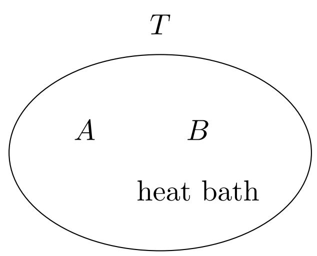


$$P(\varepsilon_i) = \frac{e^{-\beta \varepsilon_i}}{\sum_j e^{-\beta \varepsilon_j}} \qquad \text{(summed over all states } j\text{, states not levels.)}$$

The only information about the heat bath to be known is its temperature. Unlike the case of the microcanonical ensemble where $E = \bar{E}$ at eqm, the total energy of the system in thermal eqm with a heat bath is not fixed. It is fluctuating with different energy with $P(\varepsilon_i) = e^{-\beta\varepsilon_i} / \sum_j e^{-\beta \varepsilon_j}$, but the average value of $\langle \varepsilon_i \rangle$ is independent of time, because $P(\varepsilon_i)$ is independent of time.

The factor $\left(\sum_{\text{states } j} e^{-\beta \varepsilon_j} \right)$ is called 'Z', the <u>canonical partition function</u>.

And our system in thermal eqm at temp T is called a <u>"canonical ensemble."</u>

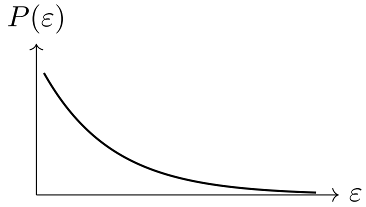


$$\boxed{Z = Z(T, V, N) = \sum_{\text{states } j} e^{-\beta \varepsilon_j} = \sum_{\text{levels } i} g_i\, e^{-\beta \varepsilon_i}}$$

The subsystem "A" in thermal eqm of this kind is not in a fixed energy state. It has got probability distributions for certain energy values. For a quantum system, "A" is not even in a pure energy eigenstate. It will be in a superposition of all energy eigenstates. *(mind blowing fact)*

$$\big(P(E) \propto e^{-\beta E}\big).$$

Distribution of energy goes like exponential decay rather than a steep Gaussian distribution.



$\beta =$ property of the bath.

In terms of energy levels,

$$Z = \sum_{\text{levels } i} g_i\, e^{-\beta \varepsilon_i} \qquad g_i \to \text{degeneracy of level } \varepsilon_i.$$

For continuous energy levels,

$$P(\varepsilon) = \frac{e^{-\beta\varepsilon}}{\displaystyle\int_0^\infty e^{-\beta\varepsilon} f(\varepsilon)\, d\varepsilon}.$$

Now, instead of counting degeneracy (g_i), we ask how many states with energy $\varepsilon \le \varepsilon + d\varepsilon$:

$$d\varepsilon \text{ (band at } \varepsilon) = f(\varepsilon)\, d\varepsilon \qquad \text{density of states}$$

$f(\varepsilon) =$ no. of states in $d\varepsilon$ range per unit energy ε.

$$Z(T, V, N) = Z = \int_0^\infty e^{-\beta\varepsilon} f(\varepsilon)\, d\varepsilon \qquad \text{(general canonical partition fun.)}$$

So,

$$P(\varepsilon) = \frac{e^{-\beta\varepsilon}}{\displaystyle\int_0^\infty e^{-\beta\varepsilon} f(\varepsilon)\, d\varepsilon}.$$

So Z is the Laplace transform of $f(\varepsilon)$, in which the transform variable is β, instead of 's'. (Laplace transform, or moment generating function, for a continuous variable.)

$$\left.\begin{cases} f(\varepsilon) \to \text{system property (mech. aspect: classical/quantum)} \\ e^{-\beta\varepsilon} \to \text{statistical aspect.} \end{cases}\right\} \qquad e^{-\beta\varepsilon} f(\varepsilon) \to \text{stat. mech.}$$

(Q) What is average energy?

$$\begin{aligned}
\langle E\rangle &= \sum_{\text{states } j} P(\varepsilon_j)\,\varepsilon_j \\
&= \frac{\sum_i \varepsilon_i\, e^{-\beta\varepsilon_i}}{Z(T,V,N)} = \frac{\sum_i \varepsilon_i\, e^{-\beta\varepsilon_i}}{\sum_j e^{-\beta\varepsilon_j}} = \frac{1}{Z}\sum_i \varepsilon_i\, e^{-\beta\varepsilon_i} \qquad (5)
\end{aligned}$$

$$\boxed{\langle E\rangle = -\frac{\partial(\ln Z)}{\partial\beta} = -\frac{1}{Z}\frac{\partial Z}{\partial\beta}}$$

$\langle E\rangle = U =$ internal energy of the system (or) $\langle H\rangle$ expectation of Hamiltonian.

(Q) Var$(E) =?$

$$\begin{aligned}
\sigma^2 &= \big\langle (E-\langle E\rangle)^2\big\rangle \\
&= \big\langle E^2 + \langle E\rangle^2 - 2E\langle E\rangle\big\rangle \\
&= \langle E^2\rangle + \langle E\rangle^2 - 2\langle E\rangle^2 \\
\sigma^2 &= \langle E^2\rangle - \langle E\rangle^2 \qquad (6)
\end{aligned}$$

$$\langle E^2\rangle = \frac{\sum_{\text{states } i} E_i^2\, e^{-\beta E_i}}{Z} = \frac{1}{Z}\frac{\partial^2 Z}{\partial\beta^2}$$

$$\sigma^2 = \frac{1}{Z}\frac{\partial^2 Z}{\partial\beta^2} - \frac{1}{Z^2}\left(\frac{\partial Z}{\partial\beta}\right)^2 = \frac{\partial^2(\ln Z)}{\partial\beta^2}$$

$$\begin{aligned}
\sigma^2 &= -\frac{\partial\langle E\rangle}{\partial\beta} \\
&= -\frac{\partial\langle E\rangle}{\partial T}\cdot\frac{\partial T}{\partial\beta} \\
&= \left(\frac{\partial\langle E\rangle}{\partial T}\right) k_B T^2 \qquad (7)
\end{aligned}$$

$$\boxed{\sigma^2 = C_V\, k_B T^2}$$

$$\left(\frac{\partial\langle E\rangle}{\partial T} = \frac{\partial U}{\partial T} = C_V\right)$$

So variance of energy is directly related to C_V.

$$\boxed{C_V = \frac{\sigma^2}{k_B T^2}}$$

So, $(C_V \geq 0)$.

In thermodynamics we only deal with $\langle E\rangle$. So we can't calculate C_V or C_P. So C_P, C_V are input in thermodynamics.

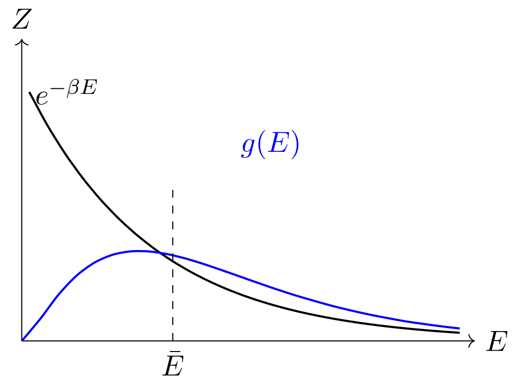
7 Lecture 25: Connection with Thermodynamics

Recall that the canonical partition function of a small subsystem in a heat bath of temp. $T = \dfrac{1}{k_B \beta}$ is

$$Z = \sum_{\text{states } j} e^{-\beta E_j} \longrightarrow \int_0^\infty dE \, e^{-\beta E} \, g(E)$$

where $g(E) \propto E^{1/2}$, $g(E) \propto E^{N/2}$ or $\propto \alpha E^N$ for $N = 10^{24}$.

Let's plot (Z vs E):



$$Z = \int_0^\infty dE \, e^{-\beta E} g(E) \, dE \;=\; e^{-\beta \bar{E}} \, \Omega(\bar{E})$$

$$(?) \text{ maybe} \sim \left(\Omega(E) \approx \Omega(\bar{E}) = \int g(E) \, dE \right) \quad \text{(by me)}$$

$$\Omega(\bar{E}) = \text{total no. of microstates}$$

So,

$$\ln Z = \ln \Omega(\bar{E}) - \beta \bar{E}$$

$$-k_B T \ln Z = -k_B T \ln \Omega(\bar{E}) + \bar{E}$$

But we know that entropy (S) is defined as

$$S = k_B \ln \Omega$$

$$= \bar{E} - TS \;=\; F \quad \text{(Helmholtz free energy)}$$

$$\Rightarrow \boxed{Z = e^{-\beta F}}$$

So, the whole idea was to write

$$Z = \sum_{\text{status } j} e^{-\beta E_j} \;\equiv\; \text{in terms of } e^{-\beta E_{\text{eff}}}, \text{ effective energy.}$$

$$(Z = e^{-\beta F}). \qquad (F = \text{Helmholtz free energy}).$$

It needs coherent punch from bath molecules to provide maximum energy to subsystem A. So it is less likely to get high energy from a heat bath; that is the reason the energy distribution goes like

$$P(E_i) = \frac{e^{-\beta E_i}}{\sum_j e^{-\beta E_j}} \qquad \text{(sum over states } j \text{ of gas)}$$

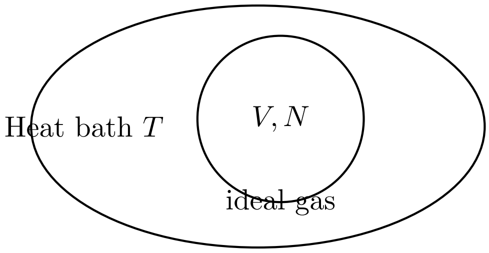
Let us find Pressure

We know that $F = U - TS$

$$\begin{aligned} dF &= dU - TdS - SdT && (8)\\ &= TdS - PdV + \mu dN - TdS - SdT && (9)\\ dF &= -SdT - PdV + \mu dN && (10) \end{aligned}$$

$$P = -\left(\frac{\partial F}{\partial V}\right)_{T,N}$$

Partition function for classical ideal gas



mass of each particle m.

Assume: All particles are independent of each other and there is no interaction b/w the particles.

$$Z = \sum_{\substack{\text{states}\\ \text{of gas}}} e^{-\beta E} = \sum e^{-\beta(\varepsilon(1)+\varepsilon(2)+\varepsilon(3)+\cdots)}$$

where $\varepsilon(i)$ is energy of i^{th} particle of whole system.

If all particles are independent of each other, then partition function factorizes into

$$Z = \sum e^{-\beta\varepsilon} = \sum \left(e^{-\beta\varepsilon}\cdot e^{-\beta\varepsilon}\cdot e^{-\beta\varepsilon}\cdots\right)$$

$$Z = \left(\sum_{\substack{\text{state of}\\ \text{any particle}}} e^{-\beta\varepsilon}\right)^N \longrightarrow \text{only true if particles are distinguishable.}$$

$$Z = \left(\int_0^\infty e^{-\beta E}\,\Omega(E)\,dE\right)^N = \left(\frac{1}{h^3}\int_V d^3x \int d^3p\, e^{-\beta\frac{p^2}{2m}}\right)^N$$

$$= \left(\frac{V}{h^3}\int e^{-\beta\frac{p_x^2}{2m}}dp_x \int e^{-\beta\frac{p_y^2}{2m}}dp_y \int e^{-\beta\frac{p_z^2}{2m}}dp_z\right)^N$$

$$= \left(\frac{V}{h^3}\Big((2\pi m k_B T)^{1/2}(2\pi m k_B T)^{1/2}(2\pi m k_B T)^{1/2}\Big)\right)^N$$

$$\boxed{Z = \left(\frac{V}{h^3}(2\pi m k_B T)^{3/2}\right)^N} \qquad \left(\int_{-\infty}^{\infty} e^{-ax^2}dx = \sqrt{\frac{\pi}{a}}\right)$$

$$Z = e^{-\beta F}, \qquad \ln Z = -\beta F \;\Rightarrow\; F = -\frac{1}{\beta}\ln Z = -k_B T\, N \ln\left(\frac{V(2\pi m k_B T)^{3/2}}{h^3}\right)$$

$$P = -\left(\frac{\partial F}{\partial V}\right)_{T,N} = \frac{1}{\beta}\cdot\frac{1}{Z}\frac{\partial Z}{\partial V}\bigg|_{T,N} = \frac{1}{\beta Z}\cdot\frac{Z}{h^3}\cdot N\left(\frac{V}{h^3}(2\pi m k_B T)^{3/2}\right)^{N-1}(2\pi m k_B T)^{3/2}$$

$$= N k_B T\cdot\frac{1}{V}\cdot\frac{(2\pi m k_B T)^{3/2}/h^3}{(2\pi m k_B T)^{3/2}/h^3}$$

$$P = \frac{N k_B T}{V} \;\Rightarrow\; \boxed{PV = N k_B T}$$

$N =$ no. of particles $\quad T =$ temp. of gas & Heat bath.

(Q) What is average energy?

$$U = \bar{E} = -\frac{\partial}{\partial\beta}\ln Z = -\frac{\partial}{\partial\beta}\left(N\ln\left(\frac{V}{h^3}\left(\frac{2\pi m}{\beta}\right)^{3/2}\right)\right)$$

$$= -N\cdot\frac{1}{\frac{V}{h^3}\left(\frac{2\pi m}{\beta}\right)^{3/2}}\cdot\frac{V}{h^3}\cdot\frac{3}{2}\left(\frac{2\pi m}{\beta}\right)^{1/2}\cdot 2\pi m\cdot\left(-\frac{1}{\beta^2}\right)$$

$$U = N\cdot\frac{3}{2}\cdot\frac{1}{\beta} = \frac{3}{2}N k_B T$$

$$\bar{E} = U = \frac{3}{2}N k_B T$$

$$\left(\frac{U}{N} = \frac{3}{2}k_B T\right) \to \text{equipartition theorem.}$$

- 3 comes from no. of dimensions
- quadratic relation $\varepsilon = \dfrac{p^2}{2m}$
  came from $\left(\int e^{-\beta\frac{p^2}{2m}}dp\right)$

$\downarrow$

Tells us that every quadratic term in 'H' (Hamiltonian) gives a contribution of $\left(\dfrac{1}{2}k_B T\right)$.

Note: not applicable for vibrational d.o.f. where the possible states [?] the particles are indistinguishable.

For $E = H = \frac{1}{2}kx^2 + \frac{p^2}{2m}$

$$\bar{E} = \frac{1}{2}k_BT + \frac{1}{2}k_BT$$
$$(\bar{E} = k_BT)$$

for $V = \frac{1}{2}kx^4$

$$H = \frac{1}{2}kx^4 + \frac{p^2}{2m}$$
$$\left(\bar{E} = \frac{k_BT}{4} + \frac{k_BT}{2} = \frac{3}{4}k_BT\right)$$

8 Lecture 25 (continued)

When we will do Q.M.; $E = pc$ for photons in black body cavity at temp T. Then $\bar{E} \neq \frac{3}{2}k_BT$ (even in 3-D).

$$\rightarrow E = pc, \text{ so}$$
$$\bar{E} \propto T^4 \qquad \text{(Bose statistics)}$$

So,

$$PV = Nk_BT; \qquad U = \frac{3}{2}Nk_BT$$
$$U = \frac{3}{2} \cdot PV$$
$$\Rightarrow \boxed{PV = \frac{2U}{3}}$$

or, $P = \frac{2U}{3}$ $\rightarrow$ avg energy density.

$$\left(u = \frac{U}{V} = \text{avg energy per unit volume}\right)$$

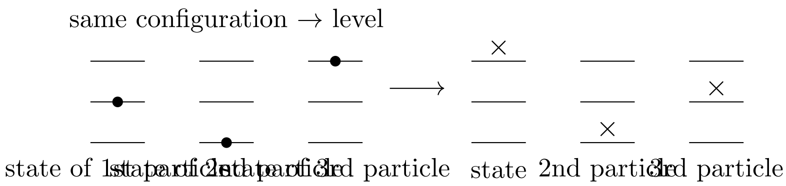
This relation is even true for quantum gases (quantum ideal gas).

(Q) Is it correct that

$$Z = \sum e^{-\beta E} = \left(\sum e^{-\beta\varepsilon}\right)^N ?$$

summed over all states of gas summed over states of any particle

No, its not true (always); here we have overcounted the number of states, since the particles are indistinguishable. (physical fact).



Since particles are indistinguishable, we have overcounted the number of states.

In quantum statistics we will correct this mistake. Instead of asking where in which level particle 1 is.. and 2 is.., we ask how many particles are in ground state, 1^{st} excited state, and so on....

So in nutshell in quantum statistics we do row-wise counting, instead of column-wise counting performed in classical statistics. From here idea of occupation no. arose in quantum statistics.

$$S = -\left(\frac{\partial F}{\partial T}\right)_{V,N} \quad ; \qquad e^{-\beta F} = Z$$

$$\left(F = -\frac{1}{\beta}\ln Z\right)$$

and,

$$Z = \left(\frac{V}{h^3}(2\pi m k_B T)^{3/2}\right)^N$$

$$\Rightarrow F = -\frac{N}{\beta}\ln\left(\frac{V}{h^3}(2\pi m k_B T)^{3/2}\right) = -Nk_BT\ln\left(\frac{V}{h^3}(2\pi m k_B T)^{3/2}\right)$$

So, $S = -\left(\dfrac{\partial F}{\partial T}\right)_{V,N}$

$$= Nk_B\ln\left(\frac{V}{h^3}(2\pi m k_B T)^{3/2}\right) + \frac{Nk_BT}{\frac{V}{h^3}(2\pi m k_B T)^{3/2}}\cdot\frac{V}{h^3}\cdot\frac{3}{2}(2\pi m k_B T)^{1/2}(2\pi m k_B)$$

$$= Nk_B\ln\left(\frac{V}{h^3}(2\pi m k_B T)^{3/2}\right) + \frac{3}{2}Nk_B$$

at $T = 0$, $\ln\left(\dfrac{V}{h^3}(0)\right) \to -\infty$, so ideal gas is no more a good assumption for our real gas, there are interactions

and physical real gas condenses to liquid & solid which also can't be explained by assuming gas to be ideal.

Also this formula of entropy is more flawed and it is due to incorrect partition function arose from overcounting of states (also known as Gibbs paradox).

Let us find

$$\mu = +\left.\frac{\partial F}{\partial N}\right|_{T,V} \quad ; \qquad F = -Nk_BT\ln\left(\frac{V}{h^3}(2\pi m k_B T)^{3/2}\right)$$

$$\mu = -k_BT\ln\left(\frac{V}{h^3}(2\pi m k_B T)^{3/2}\right)$$

This formula is also not sensible....

Even the formula for Free energy $F = -Nk_BT\ln\left(\dfrac{V}{h^3}(\ )^{3/2}\right)$

as, we expect free energy to be extensive quantity.

The 'V' in $\ln\left(\frac{V}{h^3}(\ )^{3/2}\right)$ is trouble here.

($\because F \propto$ any one extensive variable)

$$F = Nf(\text{intensive variable}).$$

Correction to partition function

There are $N!$ no. of ways particles can be arranged

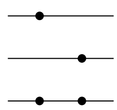
ex. 3 particles.

3	2	1
1st	2nd particle	3rd

$= 6$ ways.

So a rough way to correct Z is to divide by $(N!)$

but that is only true if no two particles attain same 'state'



But that is only very likely if no. of states available

per particle is much larger than number of particles. and indeed that happens in one case of classical ideal gas.

In general, $i, j, k, \ldots$ are energy levels of particle i.

$$Z = \frac{1}{N!}\sum_{i\neq j\neq k} + \frac{1}{2!(N-2)!}\sum_{i=j\neq k\neq l} + \frac{1}{3!(N-3)!}\sum_{(i=j=k\neq\cdots)} + \cdots$$

case if any two particles attains same level. prob. that any 3 particles have are in same level.

If number of states $(i, j, k\ldots)$ are much larger than no. of particles it is very less likely that any two or any three particles attains same state.

So,

$$Z = \frac{1}{N!}\left(\sum e^{-\beta\varepsilon}\right)^N \qquad \text{(summed over states of one particle)}$$

"This was done before advent of Q.M. by Sackur-Tetrode"

$$\left(\ln Z = \ln\frac{1}{N!}\left[\frac{V(2\pi m k_B T)^{3/2}}{h^3}\right]^N\right)$$

We know Sterling formula

$$N! \simeq N^N e^{-N} \sqrt{2\pi N} \qquad \text{(for very large } N\text{)}$$

$$N! \simeq N^N e^{-N}$$

So then, volume:

$$Z \simeq \left[\left(\frac{V}{N} \right) \frac{(2\pi m k_B T)^{3/2}}{h^3} \cdot e \right]^N$$

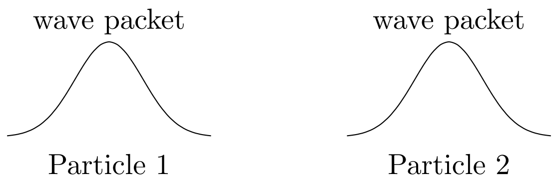
So all non-sensible formulas of (μ, S) are corrected by this. Of course $S(T = 0)$ won't be explained by ideal gas.

9 Lecture 25 (continued): Criterion for Classical Statistics

Since particles are quantum objects, we have to move to quantum statistics for more precise understanding.

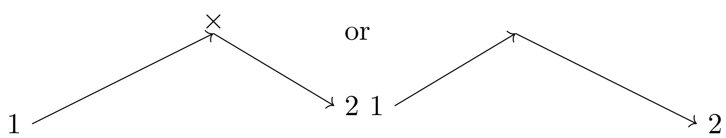
Criterion for classical statistics to be valid

In Q.M. particles are indistinguishable and x & p of particle can't be precisely defined instantaneously.



We can say that particle 1 is at this position & particle 2 is at some other position, iff the avg. distance b/w them is much larger than spread of wave packets. (i.e. non-interacting identical particles can be distinguishable).

When particles 1 & 2 interact, we can't identify them.



(particles are exchanged).

So, for classical statistics to work,

mean interparticle separation $\ggg \lambda_{\text{deBroglie}}$

we still have to divide by $(N!)$ to count states correctly.

If we have N particles in volume V then volume available for each particle is $\left(\frac{V}{N} \right)$

So linear dimension available is $\left(\frac{V}{N}\right)^{1/3}$

$$\left(\frac{V}{N}\right)^{1/3} \ggg \frac{h}{p_{rms}} \qquad \left(\lambda_{deB} = \frac{h}{p}\right)$$

$$\left\langle \frac{p^2}{2m} \right\rangle = \frac{3}{2}k_B T \quad \text{for one particle} \qquad p_{rms} = \sqrt{3mk_B T}$$

$$\sqrt{\langle p^2 \rangle} = p_{rms} \;\Rightarrow = (3mk_B T)^{1/2} \text{ or } (mk_B T)^{1/2}$$

$$\left(\frac{V}{N}\right)^{1/3} \ggg \frac{h}{(mk_B T)^{1/2}}$$

$$\left(\frac{V}{N}\right) \ggg \frac{h^3}{(mk_B T)^{3/2}} \qquad \text{or} \qquad \left(\frac{N}{V}\right) \ggg \frac{h^3}{(mk_B T)^{3/2}}$$

or

$$1 \ggg \frac{nh^3}{(mk_B T)^{3/2}} \qquad \frac{nh^3}{(mk_B T)^{3/2}} = \text{degeneracy factor}$$

So,

$$\boxed{\frac{nh^3}{(mk_B T)^{3/2}} \ll 1}$$

is criterion for statistics to be valid. (n = number density of particles)

Let's check for Nitrogen at N.T.P.: mass $m \sim 10^{-26}$kg, $h = 10^{-34}$, $n = 10^{24}$ per cubic meter, $T = 300$K, $k_B = 1.38 \times 10^{-23}$.

$$\frac{10^{24} \cdot (10^{-34})^3}{(10^{-26} \cdot 10^{-23} \cdot 300)^{3/2}} = \frac{10^{-78}}{(10^{-51})^{3/2}} \approx \frac{10^{-78}}{10^{-75}} \approx 10^{-3} \ll 1.$$

So classical statistics works.

If (n, T) this approximation fails. (like in case of liquids/neutron stars. the degeneracy factor is of order of 10^6).

Also at low temp ($T \to 0$) this limit/condition fails and we use quantum statistics.

10 Lecture 26

(Lec-26 — is of probability distributions so, It will be written in mathematical notebook of V. Balki sir).

(After) (10:00)

11 Lecture 27: Phase Transitions

Thermodynamics is a kind of limiting case of statistical mech. where fluctuations are neglected and so on; And these fluctuations can be taken into account if we use stat. mech. and it provides correction to thermodynamic cases, goes beyond thermostatics & calculate (C_P, C_V), which are input parameter in thermodynamics.

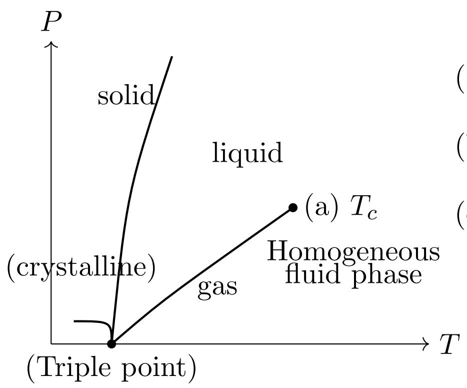
There is a place where thermodynamics fails (completely and no longer valid); it is hard problem of phase transitions of matter. In principle stat. mech. (eqm stat. mech.) should tell us everything about system at thermal eqm, including processes at phase transitions.

Thermodynamics fails at critical point (P_c, V_c, T_c).

We will use vander Waals gas to demonstrate real gases.

We have thermodynamic variables (P, V, T) out of which we only use any two of them, as third is found using equation of state (which is different for different phases of matter – solid, liquid, gas).

Phase diagram



(a) $\rightarrow$ liquid-gas coexisty curve or (Boiling curve)
(b) $\rightarrow$ solid-liquid coexisty curve or melting curve
(c) $\rightarrow$ solid-gas coexistat curve or sublimation (curve).

Every point on this phase plane is assumed to be at thermal eqm state of the system.

(phase $\rightarrow$ A homogeneous thermodynamic eqm system)

(Ice melting of glaciers is measured using rise of water level, but Ice can directly convert to vapour phase, so this is a challenging problem).

Note: It looks like the melting curve goes on with same slope, but it does not happen. As pressure increases, solid have more densely packed atoms, at very high pressure (FCC) Face centered cubic crystals forms.

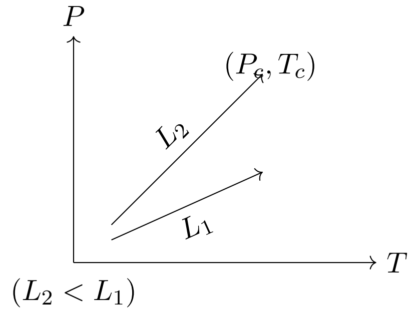
(Gibbs phase rule): For one kind of molecular species we can at most have we can have only three coexisting phases. For more than one molecular species we can have more than 3 kind of phase coexisting at common point.

Critical Point (V_c, T_c, P_c)

The boiling curve/liquid-gas coexisty curve stops at some point on phase plane called critical point where it becomes impossible to distinguish liquid phase from gas phase.

At this point the conventional thermodynamics fails. On right of (P_c, T_c) or for $(T > T_c)$ we can't tell diff. between liquid & gas phase. (Homogeneous fluid phase).

To convert liquid to gas phase, we supply latent heat. This transition is discontinuous as density of system changes discontinuously.

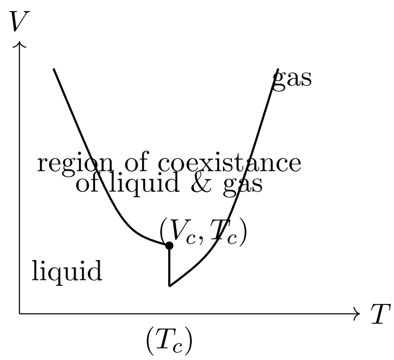


As we go up this curve (i.e. increase pressure), latent heat becomes less and less & latent heat vanishes at critical point (i.e. we require no heat to convert from liquid to gas phase).

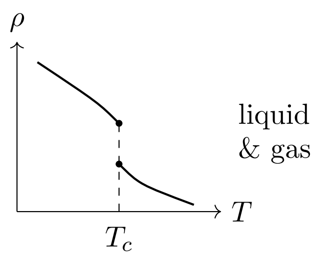
Also, the surface tension also vanishes at critical point i.e. the meniscus (the curved upper surface of liquid) also disappears. "we have mush at critical point"

Everything that distinguish liquid from gas disappears, even the density start matching.

Let's draw melting curve in the $(V - T)$ plane



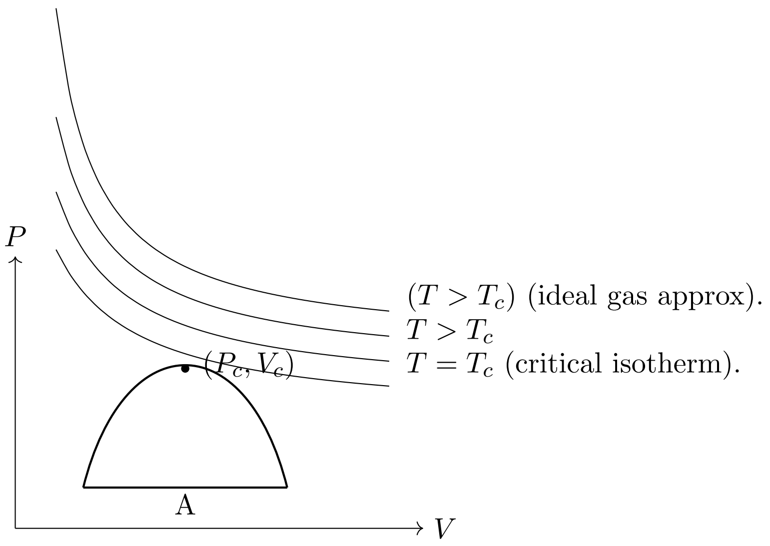
This boiling curve becomes region in V-T diagram (homogeneous fluid phase).



discontinuous phase transition jump in density (ρ).

Critical point is 2^{nd} order or continuous phase transition, and coexisting curve is series of discontinuous / 1^{st} order phase transitions.

P-V diagram (only for boiling curve)



wrong curves as $\dfrac{\partial P}{\partial V} > 0$ [?] (violates principle of stability).

$$P_c = \frac{nRT}{V - nb} - \frac{an^2}{V^2} \qquad \text{or} \qquad \boxed{\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT}$$

To find critical point (inflection point), from van der Waals equation of state,

$$\left.\frac{\partial P}{\partial V}\right|_{(T_c, V_c)} = 0$$

also

$$\left.\frac{\partial^2 P}{\partial V^2}\right|_{(P_c, V_c, T_c)} = 0$$

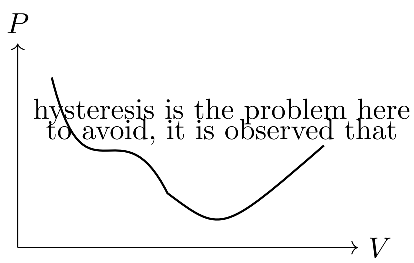
and from eq. of state, we can find (P_c, V_c, T_c).

There is no universal eq. of state as eq. of state depends upon interactions and interactions are hard to model universally.

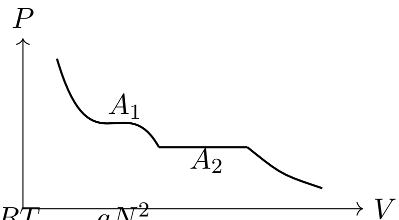
12 Lecture 27 (continued)

The portion where $\dfrac{\partial P}{\partial V}$ is positive makes compressibility negative and violates thermodynamic stability. So we have to correct this portion.

$$k_B = -\frac{1}{V}\frac{\partial V}{\partial P}$$



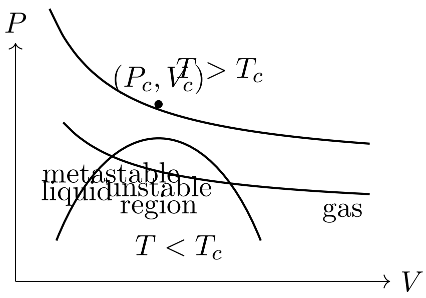
tieline transition occurs such that Area are equal (flip).



$$P = \frac{NRT}{V - Nb} - \frac{aN^2}{V^2} \text{ (tieline)}$$

So some portions of allowed curve also gets omitted in observation. So vander Waals gas eqn. of state is not accurate, we not only have to remove unstable region but slightly bigger region also.

So envelope goes like this—

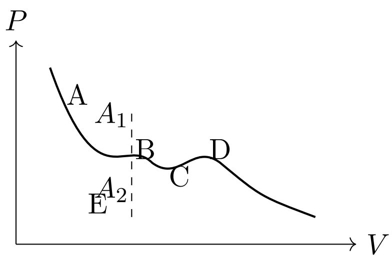


(system separates slowly into gas & liquid) (system separates to liquid & gas abruptly).

Any system on unstable region will instantaneously separate into some liquid & remaining to gas depending upon lever rule.

Note: In case of binary alloys (liquid alloys), The metastable portion can be made extremely slow, it may take hours, years to happen.

Maxwell's Equal Area Rule



(Maxwell's tieline construction) — "Isn't quite correct though as It goes through ($D \to B$), i.e. from thermodynamic unstable region."

Let us consider $G(T, P, N) = U - TS + PV$

$$dG = dU - TdS - SdT + PdV + VdP$$

Since $dU = TdS - PdV + \mu dN$,

$$dG = -SdT + VdP + \mu dN$$

On an isotherm, $dG = VdP$ (as $dT = 0$).

$$G_E - G_A = 0 = \int_A^E dG = \int_A^B VdP + \int_B^C VdP + \int_C^D VdP + \int_D^E VdP$$

$G = \mu$ should be same in gas phase or liquid phase.

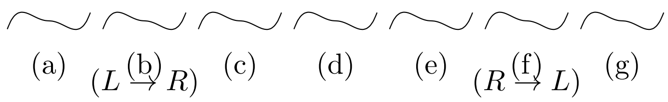
$$\left(\int_A^B VdP - \int_C^B VdP\right) + \left(-\int_C^D VdP + \int_E^D VdP\right) = 0$$

$$\underbrace{\qquad\qquad}_{(A_1)} + \underbrace{\qquad\qquad}_{(-A_2)}$$

$$\boxed{A_1 = A_2} \Rightarrow A$$

13 Lecture 27 (continued): Mechanical Example of Hysteresis

Mechanical example of hysteresis (Double well potentials).



If we start at (R) i.e. right well, Transition occurs to 'L' at (f).

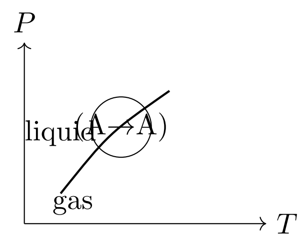
If we start at 'L' i.e. left wall (a), then transitions occur at (b). This is called hysteresis in mechanical case.

There is a branch of mathematics called catastrophe theory where people talk about change in generic potentials. (BTW) we can't have such minimum in thermal eqm.

14 Lecture 28

Near critical point, many systems behave (independent of their interactions/details) identically.

We can convert from liquid to gas or gas to liquid without encountering a sharp transitions. (This is only possible because boiling curve ends at critical point).



We can't have continuous transitions from liquid to solid or vice versa. (Because this curve (melting curve) does not end – it does not end because solid has crystalline symmetry.

It is symmetric under group of specific transformations like space rotations, translations. On the other hand liquid is homogeneous and isotropic, therefore liquid has much greater degree of symmetry. Solid has long range order.

Order and symmetry are kind of opposites. More ordered a phase, less symmetric it is; in the sense that the set of transformations under which it remains unchanged is smaller and smaller.

The most disordered state is in fact most symmetric because it looks exactly the same in every direction.

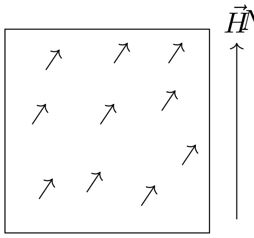
On the other hand, liquid and gas phases are both symmetric (in every dir.). So $\exists$ a critical point.

We can understand behaviour at critical point via fluid-magnet analogy.

Model of Paramagnetism

<u>Assumption</u> – system consists of elementary magnetic dipole moments (atomic (spin + angular)), oriented randomly in presence of ext. $\vec{H}_{ext}$ $\left(H = \dfrac{\vec{B}}{\mu_0}\right)$.

Magnetic dipole moment will try to align in direction of $\vec{H}_{ext}$. So system will have some kind of symmetry.



N dipole moments, $\vec{\mu}_i$

(auxiliary field)

placed in Heat bath at temp T.

P.E. of a dipole $\vec{\mu}$ in field $\vec{H}$:

$$U = -\vec{\mu} \cdot \vec{H}$$

$$U_{min} = -\mu H \quad (\theta = 0)$$

$$U_{max} = +\mu H \quad (\theta = \pi).$$

Let's look at very simple model, such that

$$U = +\mu H \text{ or } -\mu H \qquad \text{(one-dimension case).}$$

(magnetic dipole moment of e^- can only take two values), one along $\vec{H}$.

$$\begin{array}{ll} E_1 & (\mu H) \\ E_0 & (-\mu H) \end{array}$$

Let's find magnetisation (or better average magnetisation), in (1-D).

$$M = \sum_{i=1}^{N} \mu_i \quad \text{(Avg dipole moment of each dipole moment).}$$

$\big(N\langle\mu\rangle$ is only true iff all μ are independent, since we only want to see $\bar{M}$ along dir. of $\vec{H}$, let's drop vector, assuming no dipole-dipole interactions$\big)$.

$$\begin{aligned} M &= N\left(\frac{\mu\, P(H) + (-\mu)P(-H)}{P(H)+P(-H)}\right) \\ &= \frac{N\left(\mu e^{\beta\mu H} - \mu e^{-\beta\mu H}\right)}{e^{\beta\mu H} + e^{-\beta\mu H}} \qquad \left(\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}}\right) \end{aligned}$$

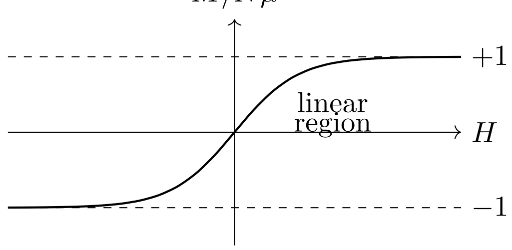
Av. magnetisation $(M) = N\mu \tanh(\beta\mu H)$

So,

$$\boxed{M = N\mu \tanh\left(\frac{\mu H}{k_B T}\right)} \quad \text{is eq. of state.}$$

fluid	(P, V, T)		$\longleftrightarrow$	magnetic (H, M, T)
field	$\longleftrightarrow$			response
(extensive)	$\longleftrightarrow$	(Intensive)		field $\longleftrightarrow$ response (extensive)

$(M/N\mu)$ **vs** H



$\left(\dfrac{M}{N\mu}\right)$ is magnetisation per atom per dipole moment.

Our old defn. of susceptibility (χ): $\vec{M} \propto \vec{H}$, $\vec{M} = \chi\vec{H}$, $\vec{M} = \text{const}\ \vec{H}$ ... is not quite right as, this graph is non linear in most range of H.

Isothermal susceptibility

$$\chi_T \equiv \left(\frac{\partial M}{\partial H}\right)_{T,N}\Bigg|_{H=0}$$

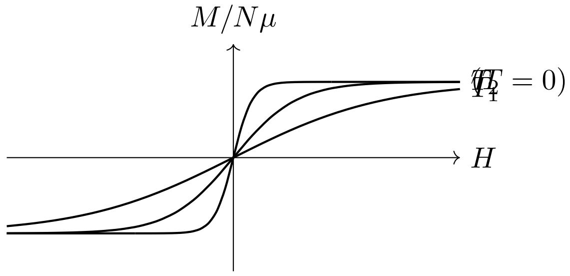
or (χ_T = slope at origin)

$$\chi_T = \left(\frac{\partial M}{\partial H}\right)_{T}\Bigg|_{H=0}$$

$\tanh(\beta\mu H) \simeq \beta\mu H$ for very small H; $\tanh(ax) \simeq ax$ for very small x.

$$\chi_T = \frac{N\mu^2}{k_B T} \rightarrow \underline{\text{Curie's law}} \quad \left(\chi_T \propto \frac{1}{T}\right) \quad \text{(Pierre Curie).}$$

So slope at origin increases with decrease in temp.



at $T = 0$ (absolute zero): $\chi_T \to \infty$, or slope $\to \infty$.

15 Lecture 28 (continued): Interactions and Ferromagnetism

We have neglected interaction between magnetic dipoles.

At high temperature, we can neglect dipole-dipole interaction but at very low temp, dipole-dipole interactions become large as compared to random fluctuation driven by heat bath.

All phase transitions of this kind is dependent on minimization of free energy

$$F = U - TS \; ; \qquad U = \langle E \rangle \text{ or } \bar{E} \text{ or } \langle H \rangle \text{ (Hamiltonian)}$$

It so happens that at sufficiently low temperatures the ordering tendency of internal energy (U) overcomes effect of entropy.

So Free energy (F) is mostly governed by U as T is very small and S (entropy) also becomes small at low temperatures $\big(S = k_B \ln \Omega(E)\big)$ ($\Omega(E)$ decreases with T).

So at low temp, F is governed by U and it is ordered phase, and at high temp F is governed by TS which is very large compared to U, and this is disordered phase.

"This is the main reason low temp is ordered state and high temp is disordered state."

If dipole-dipole interaction is taken in account our plot & relation $\left(M = N\mu \tanh\left(\frac{\mu H}{k_B T}\right)\right)$ fails.

For two dipoles, if U_d represents interaction energy:

$$\begin{matrix}\longrightarrow \\ \longrightarrow\end{matrix} \; U_{d_1} \quad \text{more stable} \quad \begin{matrix}\longrightarrow \\ \longleftarrow\end{matrix} \; U_{d_2}$$

$(U_{d_1} < U_{d_2})$

Dipole-dipole interaction is not isotropic, below configurations does not have same energy.

$$\begin{matrix} \longrightarrow & \longrightarrow & \longrightarrow & \longrightarrow \\ \longrightarrow & \longrightarrow & \longrightarrow & \longrightarrow \\ (1) & (2) & (3) & (4) \end{matrix}$$

(plane) — these configurations do not have same energy.

If we rely on dipole-dipole interactions alone, we won't have bar magnet at all. Because, one ($\uparrow \mu$) will flip another in reverse direction $(\downarrow)\mu$, so that the net dipole moment cancels out & magnetisation becomes zero.

$$(\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow\uparrow\downarrow \cdots)$$

net ($M = 0$).

But we do see permanent magnets, so the phenomena of permanent magnetism is more subtle than this and is not connected to dipole-dipole interactions,

It is connected to a quantum mechanical effect called the exchange interaction which actually favours ($\uparrow\uparrow$) over ($\uparrow\downarrow$), and it in fact overcomes dipole-dipole interactions. ($\uparrow\downarrow$).

Exchange Interaction is very strong short range interaction. It decays exponentially. (Whereas dipole-dipole interaction is long range $\left(\frac{[?]}{r^3}\right)$ & weak).

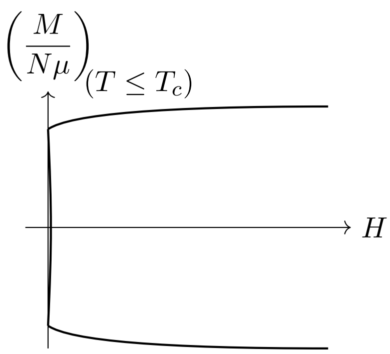
Where there are magnets where dipole-dipole interaction becomes more important/dominant and it leads to Anti-ferromagnetism:

$$(\uparrow\downarrow\uparrow\downarrow\downarrow\uparrow\downarrow\uparrow\downarrow).$$

16 Magnetism and Phase Transitions

If we take exchange interactions in account we see that the curve b/w $\left(\frac{M}{N\mu}\right)$ vs H, attains infinite slope at finite temp ($T > 0K$) instead of ($T = 0K$).

Below curie temp (T_c) the curve attains infinite slope at origin.



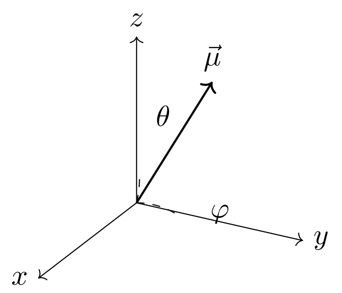
So this is discontinuous phase transitions at (T_c) curie temp.

This phase transition is from paramagnetic phase to ferromagnetic phase (M is very high even at $H = 0$).

16.1 Magnetization Calculation (General 3-D Case)

For $\rightarrow$ general case $\vec{\mu}$ at angles (θ, φ) (3-D case) — let's compute the magnetisation.

Also assuming no dipole-dipole interaction. [?]



Average magnetisation: $M = N\langle \mu \rangle$ (over solid angle)

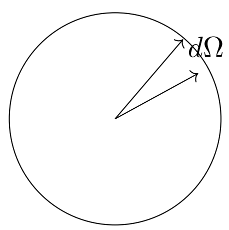
Exchange interaction (short range), energy:

$$\langle \mu \cos\theta \rangle = \frac{\displaystyle\int \mu \cos\theta \, P(\theta)\, d\Omega}{\displaystyle\int P(\theta)\, d\Omega}, \qquad \text{where } P(\theta) \propto e^{\beta H \mu \cos\theta}$$

$$\langle \mu \cos\theta \rangle = \frac{\displaystyle\int \mu \cos\theta \, e^{\beta H \mu \cos\theta}\, d\Omega}{\displaystyle\int e^{\beta H \mu \cos\theta}\, d\Omega}$$

Here we have ignored K.E., Rotational energy of these dipole moments, we have only used magnetic energy.

$$\int d\Omega = \int_{-1}^{1} d(\cos\theta) \int_0^{2\pi} d\varphi \qquad \text{or} \qquad \int d\Omega = \int_0^{\pi} \sin\theta\, d\theta \int_0^{2\pi} d\varphi$$



$$d\Omega = \frac{dA}{r^2} = \frac{(r\, d\theta)(r \sin\theta\, d\varphi)}{r^2} = \sin\theta\, d\theta\, d\varphi$$

$$\int d\Omega = \iint \sin\theta\, d\theta\, d\varphi$$

$$M = N\langle \mu \rangle = N\mu \cdot \frac{\displaystyle\int_0^{\pi} \cos\theta\, e^{\beta H \mu \cos\theta} \sin\theta\, d\theta \int_0^{2\pi} d\varphi}{\displaystyle\int_0^{\pi} \sin\theta\, d\theta \int_0^{2\pi} d\varphi}$$

Let $t = \cos\theta \Rightarrow dt = -\sin\theta\, d\theta$:

$$= \frac{N\mu(2\pi) \displaystyle\int_{-1}^{1} t\, e^{\beta H \mu t}\, dt}{2\pi \displaystyle\int_0^{\pi} e^{\beta H \mu \cos\theta} \sin\theta\, d\theta} = \frac{2\pi N\mu \displaystyle\int_{-1}^{1} t\, e^{\beta H \mu t}\, dt}{2\pi \displaystyle\int_{-1}^{1} e^{\beta H \mu t}\, dt}$$

$$= N\mu \left(\frac{t\, e^{\beta H\mu t}}{\beta H\mu} - \frac{e^{\beta H\mu t}}{\beta^2 H^2 \mu^2} \right) \Bigg|_{-1}^{1} \Bigg/ \left(\frac{1}{\beta H\mu} \right) \left(e^{\beta H\mu t} \right) \Big|_{-1}^{1}$$

$$= \frac{N\mu^2 H\beta \left(e^{\beta H\mu} \left(\frac{1}{\beta H\mu} - \frac{1}{\beta^2\mu^2H^2} \right) - e^{-\beta H\mu} \left(\frac{-1}{\beta H\mu} - \frac{1}{\beta^2\mu^2H^2} \right) \right)}{\left(e^{\beta H\mu} - e^{-\beta H\mu} \right)}$$

$$= N\mu \left[\frac{e^{\beta H\mu} + e^{-\beta H\mu}}{e^{\beta H\mu} - e^{-\beta H\mu}} - \frac{1}{\beta H\mu} \right]$$

$$\boxed{M = N\mu \left[\coth(\beta H\mu) - \frac{1}{\beta H\mu} \right]}$$

16.2 Langevin Function and Susceptibility

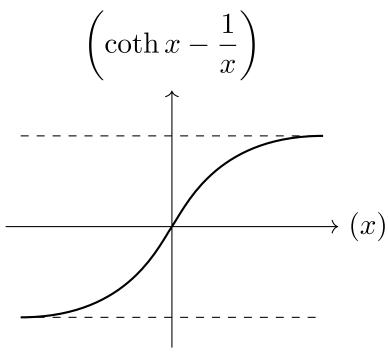
Previously when we only assumed $\mathcal{E} = \pm\mu H$:

$$M = N\mu \tanh x, \qquad x = \beta\mu H$$

Now when we used all possible orientations

$$M = N\mu \left(\coth x - \frac{1}{x} \right), \qquad x = \beta\mu H \qquad \text{(Langevin function)}$$

Let us plot $\left(\frac{M}{N\mu} \right)$ vs x, and $\left(\coth x - \frac{1}{x} \right)$ vs (x):



$$\coth x = \frac{\cosh x}{\sinh x} = \frac{1 + \frac{x^2}{2} + \cdots}{x \left(1 + \frac{x^2}{6} + \cdots \right)} = \frac{1}{x} \left(1 + \frac{x^2}{2} + \cdots \right) \left(1 - \frac{x^2}{6} + \cdots \right)$$

$$= \frac{1}{x} \left(1 + x^2 \left(\frac{1}{2} - \frac{1}{6} \right) + \cdots \right) = \frac{1}{x} \left(1 + \frac{x^2}{3} \right)$$

$$\coth x \simeq \frac{1}{x} + \frac{x}{3}$$

$$\left(\coth x - \frac{1}{x}\right) \simeq \frac{x}{3}$$

So,

$$\frac{M}{N\mu} = \left(\coth\frac{\mu H}{k_B T} - \frac{k_B T}{\mu H}\right)$$

So, near origin ($H = 0$):

$$\left.\frac{M}{N\mu}\right|_{H\to 0} \simeq \frac{\mu H}{3k_B T}$$

So

$$\chi_T = \frac{N\mu^2}{3k_B T} \qquad (\chi_T = \text{Isothermal susceptibility}), \qquad \text{No. of dimensions is '3' in one case.}$$

16.3 Two-Dimensional Case

(For 2-D) :- μ is allowed to have any direction in a plane.

$$\frac{M}{N\mu} = \frac{N\displaystyle\int_0^{2\pi} \mu\cos\theta\, e^{\beta H\mu\cos\theta}\, d\theta}{\displaystyle\int_0^{2\pi} e^{\beta H\mu\cos\theta}\, d\theta}$$

(This integral is related to the modified Bessel function I_0; not elementary.)

Since we are only interested in finding χ_T, i.e. slope at origin, let's expand $e^{\beta H\mu\cos\theta}$ to first order:

$$\frac{M}{N\mu} = \frac{\displaystyle\int_0^{2\pi} \cos\theta\left(1 + \beta H\mu\cos\theta + \frac{(\beta H\mu\cos\theta)^2}{2!} + \cdots\right) d\theta}{\displaystyle\int_0^{2\pi} \left(1 + \beta H\mu\cos\theta + \frac{(\beta H\mu\cos\theta)^2}{2!} + \cdots\right) d\theta}$$

We are only interested in finding (χ_T) i.e. slope at origin. So lets expand $\left(e^{\beta H\mu\cos\theta}\right)$ to first order.

$$\frac{M}{N\mu} = \frac{0 + \beta H\mu\displaystyle\int_0^{2\pi}\cos^2\theta\, d\theta}{2\pi + \beta H\mu\left(\displaystyle\int_0^{2\pi}\cos\theta\, d\theta\right)^{\to 0}} = \frac{\beta H\mu\,\pi}{2\pi}$$

$$= \frac{\beta H\mu}{2}$$

So slope

$$\left.\frac{\partial M}{\partial H}\right|_{H=0} = \frac{N\beta\mu^2}{2} = \frac{\mu^2 N}{2k_B T} \qquad \text{(dimension is 2)}$$

$$\boxed{\chi_T = \frac{N\mu^2}{2k_B T}}$$

What we saw that $\left(\chi_T \propto \dfrac{1}{T}\right)$ but it is not valid at small temp ($T \to 0$ or $T = 0$). So we need to fix it.

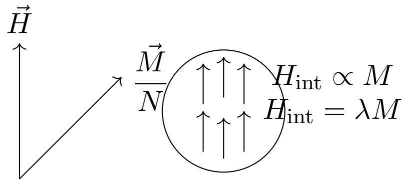
16.4 Weiss Molecular Field Theory

Just like to go from ideal gas to real gas (use interactions) we used the van der Waals gas eq^n, similarly here, we use Weiss molecular field theory.

Weiss molecular field theory

The idea is that in the ferromagnetic medium each dipole moment experiences not only the ext. field H but also the internal mag. field due to other dipole moments.

replace $H \to H_{\text{eff}} = H + \dfrac{\lambda M}{N}$ (to make intensive quantity to add to H)



$$M = N\mu \tanh\left(\frac{\mu\left(H + \frac{\lambda M}{N}\right)}{k_B T}\right) \qquad \text{(for } T \to 0\text{)}$$

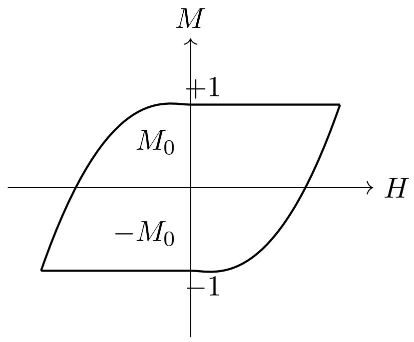
This is transcendental equation, we can only solve it by numerical methods.

Let us see its behaviour near $H = 0$.

$$M_0 = N\mu \tanh\left(\frac{\mu\lambda M_0}{N k_B T}\right) \qquad (M_0 = \text{magnetisation per unit dipole moment})$$

(1) $M_0 = 0$ is always a solution (Paramagnetic solution). ($H = 0, M = 0$)

(2) Real ferromagnets have hysteresis.

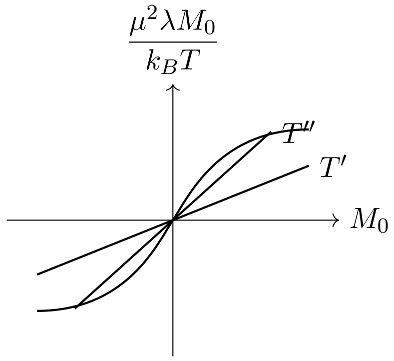


[?] (slope/intercept annotation near origin of the hysteresis loop is unclear)

Near $M \geq 0$:

$$M_0 = N\mu \tanh\left(\frac{\mu\lambda M_0}{Nk_BT}\right) \simeq \frac{N\mu^2\lambda M_0}{Nk_BT} \simeq \frac{\mu^2\lambda M_0}{k_BT}$$

Let's plot $\left(\frac{\mu^2\lambda M_0}{k_BT}\right)$ vs (M_0), i.e. (at high temp $T' > T_c$, slope is less), (for smaller temp T'' slope is even higher).



Intersection happens if (slope of line) > 1:

$$\frac{\mu^2\lambda}{k_BT} > 1, \qquad \text{or} \qquad \boxed{T < \frac{\mu^2\lambda}{k_B}}$$

So $T = \dfrac{\mu^2\lambda}{k_B}$ is critical point.

The negative solution for M_0 is unstable; so we only have one positive root of M_0, which leads to ferromagnetism.

So below a certain critical temp, the cooperative tendency of all these magnetic moments to align in same dir. will dominate over disrupting tendency of entropy (S), and we have order.

$$F = U - TS \qquad (U \gg TS)$$

at low temp (T): $F \approx U$ (ordered phase) (ferromagnetic)

but above (T_c), disorder wins and interaction is not strong enough to maintain order.

$$F = U - TS \qquad (TS \gg U)$$

This is how phase transition occurs b/w paramagnet and ferromagnet. Now we will see how it is similar to case of fluids.

17 Lecture 29: Ferroelectric Transitions and the Critical Region

We can also have ferroelectric transitions of permanent electric dipole moments, which are accompanied by structural phase transitions (crystalline phase transitions) which would lead to shapes of unit cells which have permanent electric dipole moments.

The behaviour near the critical region of eq^n of state

$$M = N\mu \tanh\left(\mu\left(\frac{H + \lambda M}{N}\right) \Big/ k_BT\right)$$

would be very similar to that of fluid near its critical region.

[?] (a boxed condition, likely of the form $\lambda N/k_BT \gg 1$, is only partially legible)

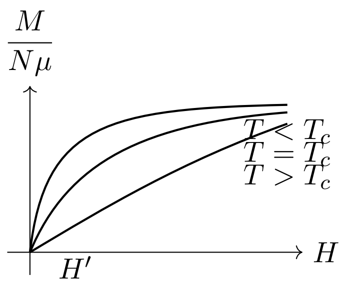
We would like to find out the (non-zero) magnetisation (M) for $H = 0$.

$$M_0 = N\mu \tanh\left(\frac{\mu\lambda M_0}{Nk_BT}\right)$$

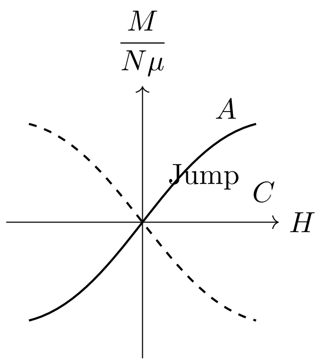
17.1 Digression: Hysteresis and Domains

Also our model $M = N\mu \tanh\left(\frac{\mu(H+\lambda M/N)}{k_BT}\right)$ does not include hysteresis phenomena. Hysteresis actually arose from presence of multiple domains in magnetic material. These domains align in right direction in presence of strong H. For a single domain our model works perfectly; a single domain can't explain/show hysteresis.

17.2 Family of Isotherms

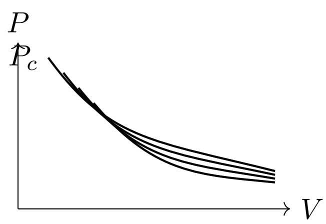


Critical Isotherm: $\left.\frac{\partial M}{\partial H}\right|_{T_c} = 0$, $\left.\frac{\partial^2 M}{\partial H^2}\right|_{T_c} = 0$.



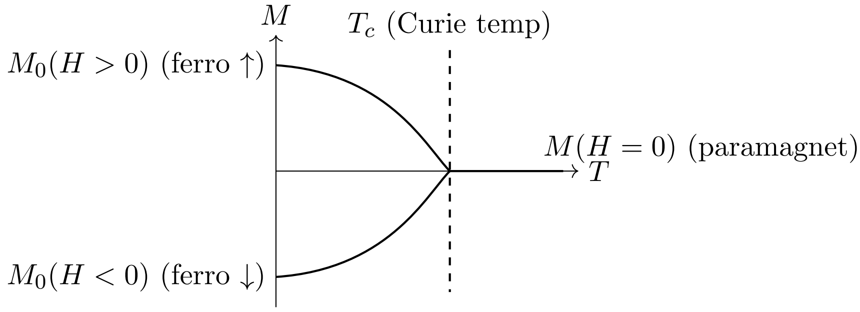
Curve $A \to C$ is unphysical, as ($H \uparrow$ in $-$ve direction, $\lambda M \uparrow$ in $+$ve direction).

Isotherms in case of fluid:



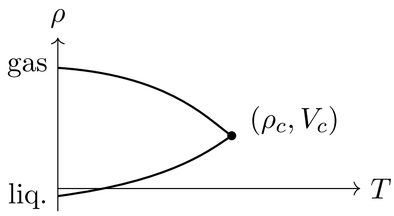
Critical Isotherm: $\left.\dfrac{\partial P}{\partial V}\right|_{(P_c,V_c)} = 0, \ \left.\dfrac{\partial^2 P}{\partial V^2}\right|_{(P_c,V_c)} = 0.$

Let us plot (M vs T):



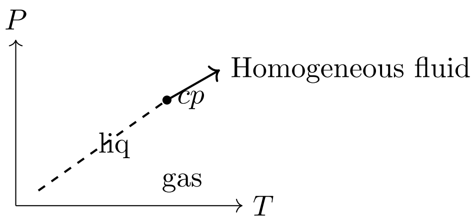
In case of (magnet): discontinuous transitions of M_0 at T_c.

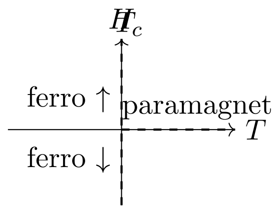
In case of fluid:



Discontinuous transitions, parabolic region near critical point.

17.3 Analogy with the Fluid Critical Point





In case of fluid as well as ferromagnet, we have one line ending at critical point.

$$\frac{dP}{dT} = \frac{L}{T\Delta v} = \frac{T\Delta S}{T\Delta v} = \frac{\Delta S}{\Delta v} = \frac{S_{\text{gas}} - S_{\text{liq}}}{v_{\text{gas}} - v_{\text{liq}}} \qquad \text{(specific volume)}$$

$$\frac{dH}{dT} = \frac{\Delta S}{\Delta M}$$

↑↑↑ has same entropy as ↓↓↓

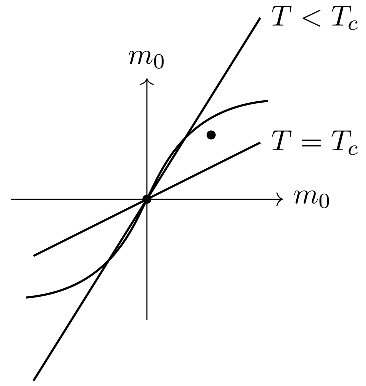
$$\left(\frac{dH}{dT} = \frac{\Delta S}{\Delta M} = \frac{0}{\text{nonzero}} = 0\right) \checkmark$$

17.4 Curie Temperature from the Mean-Field Equation

Let's find $M_0 = N\mu \tanh\left(\dfrac{\mu\lambda M_0}{Nk_BT}\right)$.

Let $\left(\dfrac{M_0}{N\mu} \equiv m_0\right)$:

$$m_0 = \tanh\left(\frac{\mu^2\lambda m_0}{k_BT}\right)$$



T_c is given by when slope is 45°:

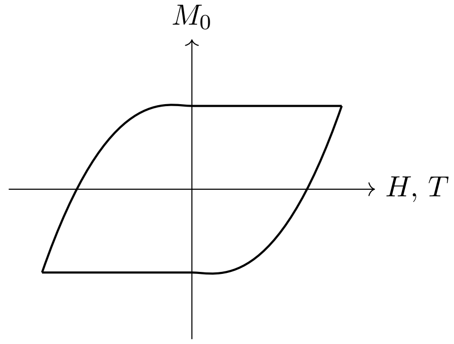
$$\left(\frac{\mu^2\lambda}{k_BT} = 1\right) \qquad \Rightarrow \qquad \boxed{T_c = \frac{\mu^2\lambda}{k_B}}$$

$(M_0 = 0)$ root is not stable for $(T < T_c)$ as magnet becomes ferromagnet instead of paramagnet. We can show it using $F = U - TS$ $(T < T_c) \to$ ordered phase.

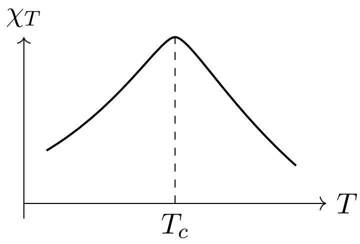
17.5 Susceptibility at $H = 0$

$$\chi_T = \frac{\partial M}{\partial H}\bigg|_T\bigg|_{H=0}$$

Even though (M) is discontinuous we can define χ_T as slope, are continuous.



Let's plot $(\chi_T$ vs $T)$:



$$m_0 = \tanh\left(\frac{\mu^2\lambda(m_0)}{k_BT}\right) = \tanh\left(\frac{T_c m_0}{T}\right)$$

at $T = T_c$: $m_0 = \tanh(m_0) \Rightarrow (m_0 = 0)$ is only solution (paramagnet).

but below $(T < T_c)$ we have other solutions.

Let us expand $\tanh x$, near [origin]:

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{\left(1 + x + \frac{x^2}{2!} + \cdots\right) - \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots\right)}{\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}\right) + \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!}\right)}$$

$$= \frac{2\left(x + \frac{x^3}{3!}\right)}{2\left(1 + \frac{x^2}{2!}\right)} = \left(x + \frac{x^3}{3!} + \cdots\right)\left(1 - \frac{x^2}{2!} + \cdots\right)$$

$$\tanh x \simeq x + x^3\left(\frac{1}{6} - \frac{1}{2}\right) = x - \frac{x^3}{3} + \cdots$$

17.6 Critical Exponent β

$$m_0 = \tanh\left(\frac{T_c}{T} m_0\right)$$

$$m_0 \simeq \frac{T_c}{T} m_0 - \frac{1}{3}\left(\frac{T_c}{T} m_0\right)^3$$

for m_0 other than zero,

$$m_0\left(1 - \frac{T_c}{T}\right) \simeq -\frac{1}{3}\left(\frac{T_c}{T}\right)^3 m_0^2$$

$$1 - \frac{T_c}{T} \simeq -\frac{1}{3}\left(\frac{T_c}{T}\right)^3 m_0^2$$

$$\left(\frac{T_c - T}{T}\right) \simeq \frac{1}{3}\left(\frac{T_c}{T}\right)^3 m_0^2$$

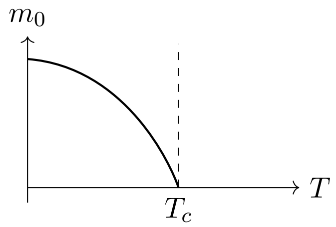
(mean field critical exponent (β); here $\beta \neq \frac{1}{k_BT}$, just a symbol)

$(T \lesssim T_c)$: if T is very near to T_c $\left(\frac{T_c}{T} \approx 1\right)$

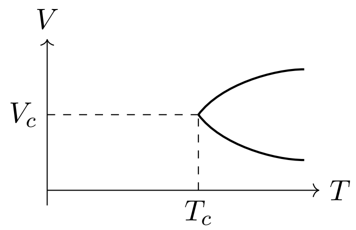
$$\Rightarrow \qquad m_0 \sim \pm\sqrt{T_c - T} \qquad \text{i.e.} \quad m_0 \simeq \pm(T_c - T)^{1/2}$$

$$\left.\frac{dm_0}{dT}\right|_{T=T_c} = \frac{\pm(-1)}{2\sqrt{T_c - T}} \to \infty$$

So behaviour of m_0 near T_c is:



Exactly same relation arose in case of fluid: $(v - v_c) \sim \pm\sqrt{T_c - T}$.



(H.W.) Show that $(v - v_c) \sim \pm\sqrt{T_c - T}$.

We can see that in both cases critical exponent is $\left(\frac{1}{2}\right)$, which arose because both are examples of mean field theory models, i.e. in van der Waals case we assume $P \neq \frac{RT}{v-b}$ but $P = \frac{RT}{v-b} - \frac{a}{v^2}$ because of attraction of each molecule. In same way here we assume every magnetic moment [feels] another magnetic moment and internal H is proportional to magnetisation itself. We have exactly the same philosophy, and that is what led to critical exponent.

(Q) Does experiment agree with critical exponent to be (1/2), or how accurately we can get closer to T_c and control temp. to see it.

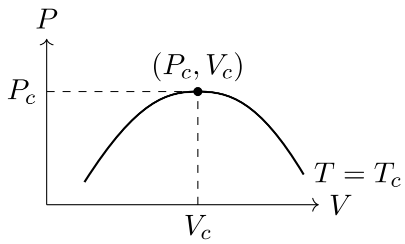
<u>Ans</u> $\sim$ In most cases we essentially see exponent closer to 1/2, but, but, if we do careful experiments near T_c we do not see 'half' at all. We see (1/3 or 0.3) for it; we have to get very-very close to T_c to see it.

Temp is one of hardest things to control; best we can control is of milli-degree of accuracy.

Frequency is most accurately measurable quantity, we can measure frequency in one part in 10^{15}. There are astronomical objects/clocks (pulsars) whose rate of change of time period can be computed to one part in 10^{19}.

17.7 Critical Isotherm and Exponent δ

(Q) How does critical Isotherm behave? (near (V_c, P_c)).



$$\left.\frac{\partial P}{\partial V}\right|_{T_c} = \left.\frac{\partial^2 P}{\partial V^2}\right|_{T_c} = 0$$

Shifting origin to see behaviour near (V_c, P_c):

$$\tilde{P} = \frac{P - P_c}{P_c}, \qquad \tilde{V} = \frac{V - V_c}{V_c}$$

The curve near (V_c, P_c) is like: (mean field critical exponent δ)

$$\left(\tilde{P} \sim |\tilde{V}|^{\delta}\right) \qquad \text{or} \qquad \left(|\tilde{V}| \sim \tilde{P}^{1/3}\right)$$

Let's look at critical isotherm for ferromagnetic transitions:

$$M = N\mu \tanh\left(\mu\left(H + \frac{\lambda M}{N}\right)\Big/ k_B T\right)$$

Let $m = \dfrac{M}{N\mu}$:

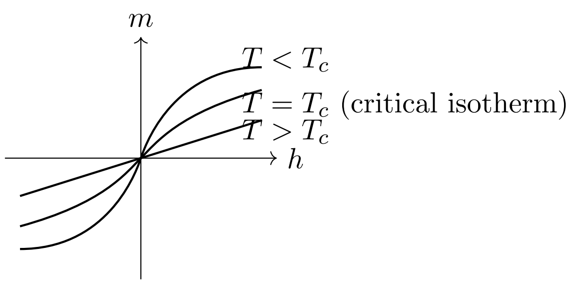
$$m = \tanh\left(\frac{\mu}{k_B T}(H + \lambda\mu m)\right) = \tanh\left(\frac{\lambda\mu^2}{k_B T}\left(\frac{H}{\lambda\mu} + m\right)\right)$$

Let $\left(\dfrac{H}{\lambda\mu} = h\right)$:

$$= \tanh\left(\frac{T_c}{T}(h + m)\right)$$

For critical isotherm ($T = T_c$):

$$\boxed{m = \tanh(h + m)}$$



at $(h = 0), (m = 0)$: m is zero (for $T \geq T_c$).

$$m = \tanh(h + m) \simeq (h + m) - \frac{(h + m)^3}{3}$$

$$\Rightarrow \qquad (m + h)^3 \sim h$$

h is much smaller (near origin), so $\underline{m^3 \sim h}$.

17.8 Susceptibility Exponent γ and Landau Theory

$$\boxed{m \sim h^{1/3}} \qquad (h \text{ is field variable})$$

Just like $\left(|\tilde{V}| \sim \tilde{P}^{1/3}\right)$ ($\tilde{P}$ is field variable).

$(m, \tilde{V})$ are response variables.

Similarly, χ_T and compressibility behave exactly same near critical point.

$$\chi_T = \left.\left.\frac{\partial M}{\partial H}\right|_T\right|_{H=0}; \qquad \kappa_T = -\frac{1}{v}\frac{\partial v}{\partial P}, \qquad \left.\frac{\partial P}{\partial v}\right|_{\text{crit. pt}} = 0$$

(Curie Weiss law) $\longleftrightarrow$

$$\chi_T = \frac{1}{|T - T_c|}$$

$$\left(\chi_T\big|_{T=T_c} = \infty\right) \quad \longleftrightarrow \quad (\kappa_T = \infty)$$

$$\chi_T = \frac{1}{|T - T_c|^\gamma} \quad \to \gamma \text{ critical point} \qquad (\gamma \neq C_P/C_V, \text{ just a symbol})$$

Experimental values of ($\gamma \sim 1.3$).

All these phase transitions can be explained by Landau's theory (1937). (Idea of broken symmetry leads to such phase transitions of all kinds of systems).